EntroDiff: Taming Hyperbolic Shocks via Double-Burgers Coupling

Anonymous Author(s)
Affiliation
Address
email

Abstract

Generative diffusion in function space has achieved striking accuracy on smooth PDE benchmarks, yet hyperbolic conservation laws remain untouched—the empirical, analytical, and algorithmic blind spot of the paradigm. Central-difference evaluation of PDE-residual guidance terms, the workhorse of the current state of the art, becomes singular at the discontinuity interfaces that define the hyperbolic regime. We resolve this impasse by observing that the score field of a VE-SDE diffusion model itself satisfies a viscous Burgers equation—a recent identity that, when the target PDE is also of Burgers type, creates two coupled Burgers structures in orthogonal directions. Physical shocks determine the singular support of the data distribution and thereby pin the score-level interfacial layer, forming a geometric correspondence that we formalise as a double-Burgers coupling theorem. Converting this structural observation into an algorithm, we embed the exact $\tanh$ interfacial profile of the score field directly into the network architecture, decomposing the learned score into a smooth background potential, a signed-distance-to-shock function, and a Rankine–Hugoniot–admissible jump amplitude. This hard-coded architectural prior eliminates the $\exp(\Lambda T_d)$ error-amplification factor that plagues standard denoisers near shocks, degrading the Gronwall constant in the convergence analysis from Λ to $\mathcal{O}(1)$. The resulting sampler, *EntroDiff*, carries a composite loss family in which each regulariser targets a distinct hypothesis in the convergence proof, and a Godunov-form PDE guidance term—entropy-consistent and well-defined across arbitrary jumps—replaces the singular central-difference residual of prior work. Under the BV-aware parameterization and a total-variation penalty that enforces L^1-compactness, we prove that generated samples converge to the Kruzhkov entropy solution at rate $W_1 \leq \mathcal{O}(\varepsilon^{1/2})$ in the score-matching error budget ε^2, with constants independent of the amplification exponent. Toy experiments on the inviscid Burgers equation validate the mechanism: the architectural prior enables effective sampling with substantially fewer integration steps than the baseline, and each ablation removes a measurable piece of the empirical margin.

1 Introduction

Generative diffusion in function space is reshaping data-driven PDE solving. The current frontier—DiffusionPDE [7] and FunDPS [18] in Banach spaces—achieves striking accuracy on Darcy, Helmholtz, and moderate-Reynolds Navier–Stokes. Yet hyperbolic conservation laws remain untouched. The avoidance is methodological, not accidental: this class is the simultaneous *empirical*, *analytical*, and *algorithmic* blind spot of the paradigm.

Empirically, the five PDE benchmarks of Yao et al. [18] are every one elliptic or parabolic; the sole Burgers experiment of Huang et al. [7] fixes $\nu_{\text{phys}} = 10^{-2}$ to sidestep the inviscid limit.

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Analytically, existing convergence theory for diffusion solvers operates in L^2, W_2, or TV under classical smoothness; none accommodates weak solutions, L^1-contraction, or the Kruzhkov entropy condition [10]. *Algorithmically*, the PDE-residual guidance central to both DiffusionPDE and FunDPS evaluates spatial derivatives by central differences—a discretisation that is singular at discontinuity interfaces. The guidance velocity itself diverges at the very feature that defines the hyperbolic regime.

A recent identity changes the calculus. Sarkar [14] establish that the score field of a VE-SDE diffusion model satisfies a viscous Burgers equation along the diffusion-time axis: $\partial_\tau s + 2s \cdot \nabla_u s = \Delta_u s$. Equivalently, $-2s$ is itself a Burgers velocity field. When the target distribution consists of solutions to a hyperbolic conservation law—itself of Burgers type in space-time—two Burgers structures coexist in orthogonal directions, and they are *coupled*: physical shocks define the singular support of the data law, thereby fixing the location of the score-level interfacial layer through the correspondence $\Sigma_{\mathrm{phys}}(t) \leftrightarrow \Sigma_{\mathrm{score}}(\tau)$ (Figure 2, Theorem 1).

This coupling reframes shock geometry from an obstacle into an asset. In standard diffusion training, a neural denoiser must *learn* the sharp $\tanh$ profile of the score near a shock from finite data; the approximation error there is amplified by an $\exp(\Lambda T_d)$ factor [14, Theorem 6.3], rendering the shock region fragile. The double-Burgers picture suggests a different route: *hard-code* the exact $\tanh$ interfacial profile into the network architecture, reducing the learning task from reproducing a singular function to merely locating it and estimating its amplitude.

The following three contributions execute this programme and provide its theoretical certification (quantitative validation in §5):

- **(C1) Double-Burgers coupling (Theorem 1).** We formalise the geometric correspondence between $\Sigma_{\mathrm{phys}}(t)$ and $\Sigma_{\mathrm{score}}(\tau)$ as a coupled Burgers system on (τ, t), establishing the structural foundation.
- **(C2) BV-aware score parameterization (§3).** The exact $\tanh$ interfacial profile of the score field [14, Prop. 5.4] is embedded into the network as an architectural prior: $s_\theta = \nabla\phi_\theta^{\mathrm{sm}} + (\kappa_\theta/2)\,\tanh(\phi_\theta^{\mathrm{sh}}/2\sigma(\tau)^2)\,\nabla\phi_\theta^{\mathrm{sh}}$. The network learns only the shock location ($\phi_\theta^{\mathrm{sh}}$) and amplitude ($\kappa_\theta$), not the shock shape.
- **(C3) Improved rate $W_1 \le \mathcal{O}(\varepsilon^{1/2})$ (Theorem 3).** Under (C2) and a BV regulariser, generated samples converge to the Kruzhkov entropy solution at rate $\varepsilon^{1/2}$ in the score-matching error, with constants *independent* of the Score Shocks amplification exponent Λ.

2 Related Work and Background

We situate the paper in four research threads and provide minimal background for the hyperbolic and metric tools used throughout.

Function-space diffusion. Lim et al. [12] pioneered score-based diffusion on Banach spaces. Huang et al. [7] introduced PDE-residual–guided sampling for forward and inverse problems. Yao et al. [18] extended Tweedie sampling to infinite-dimensional Banach spaces, reaching 3% observation density on five benchmarks—all of which are elliptic or parabolic. The common denominator is a PDE-residual term evaluated by central differences. **Where we differ:** we target *hyperbolic* conservation laws, for which central differences are singular at shock interfaces. Our Godunov-form flux (8) replaces them, remaining well-defined across arbitrary jumps.

Flow matching for PDEs. Stochastic interpolants [2] and their function-space lifts [4] unify flow- and diffusion-based solvers. These methods engineer the generative dynamics; we retain the VE-SDE backbone and manipulate the *intrinsic* structure of its score field—the Burgers identity (9) that persists independently of the chosen transport path.

The PDE perspective on diffusion. Sarkar [14] established that the score field of a VE-SDE satisfies a viscous Burgers equation, derived the exact $\tanh$ shock profile (Proposition 5.4), and identified an $\exp(\Lambda T_d)$ error-amplification factor (Theorem 6.3) that is sharp under standard parameterizations. Armegioiu [3] bounded Wasserstein stability for chaotic PDE measure transport via average-strain exponents. **Where we differ:** we convert the score–Burgers identity from an analytical observation into both an architectural prior (4) and a convergence rate (12) that removes the $\exp(\Lambda T_d)$ exponent.

Hyperbolic conservation laws. Classical solvers—WENO [15], discontinuous Galerkin [5], and Godunov-type schemes—serve here as ground-truth generators. The Kruzhkov [10] and Kuznetsov [11] theories provide the L^1-contractive semigroup and the vanishing-viscosity rate that scaffold Theorems 2–3.

2.1 Background

VE-SDE preliminaries. We adopt the EDM formulation [9]. Forward: inject Gaussian noise of variance $\sigma^2(\tau)$ at diffusion time $\tau \in [0, T_d]$, yielding noised density $p_\tau = \rho^\star * G_\tau$ with score $s(u,\tau) = \nabla_u \log p_\tau(u)$. The forward evolution obeys $\partial_\tau p_\tau = \Delta_u p_\tau$, from which (9) follows via Cole–Hopf (Theorem 1). The reverse SDE $du = -2\dot\sigma^2(\tau)\, s(u,\tau)\, d\tau + \sqrt{2\dot\sigma^2(\tau)}\, d\bar W_\tau$ is trained by minimising $\mathcal{L}_{\mathrm{DSM}}$ of s_θ. The final sample at $\tau = 0$ is u^θ, with $\mathrm{Law}(u^\theta)$ its distribution.

Hyperbolic target. The target scalar conservation law is

$$\partial_t \mathbf{u} + \partial_x f(\mathbf{u}) = 0, \tag{1}$$

with C^1 flux (canonical: Burgers, $f(u) = u^2/2$). Solutions may break down in finite time; the unique Kruzhkov entropy solution $\mathbf{u}^\star$ [10] is selected by the entropy condition, and its semigroup is L^1-contractive—the analytical backbone of §4.

Metrics. Distributional error is measured by W_1 (Kantorovich–Rubinstein dual); solution regularity is controlled by $\mathrm{TV}(\cdot)$. The compact embedding $\mathrm{BV}(\Omega) \hookrightarrow L^1(\Omega)$ (Helly) and the equivalence of L^1 and W_1 on bounded-TV sets provide the compactness used in Theorem 3.

3 Method: *EntroDiff*

Theorem 1 establishes that the physical shock set $\Sigma_{\mathrm{phys}}(t)$ determines the score-level interfacial layer $\Sigma_{\mathrm{score}}(\tau)$ through the data distribution ρ_t. This structural coupling is the algorithmic principle of the paper: rather than asking a neural network to *learn* shock geometry from finite data—an ill-conditioned problem whose approximation error is amplified by $\exp(\Lambda T_d)$ [14]—we *embed* the known analytic structure of shocks into the architecture, the loss, and the sampler. Four coordinated design choices (§3.1–3.4) instantiate this principle; their joint presence is necessary for Theorem 3 to excise the exponential factor.

3.1 Viscosity-matched noise schedule

The score-level Burgers equation (9) carries effective viscosity $g(\tau)^2/2$ with $g(\tau) = \sqrt{\dot\sigma^2(\tau)}$ for a VE-SDE. The solution-level conservation law (1), in its vanishing-viscosity limit, is the $\nu_{\mathrm{phys}} \to 0^+$ limit of $\partial_t \mathbf{u} + \partial_x f(\mathbf{u}) = \nu_{\mathrm{phys}}\, \partial_{xx}\mathbf{u}$. Theorem 1 achieves its tightest geometric correspondence when the two viscous scales coincide. We therefore replace the default EDM schedule with

$$\sigma^2(\tau) = 2\,\nu_{\mathrm{phys}}\,\tau, \qquad \tau \in [0, T_d]. \tag{2}$$

Under (2), the forward SDE is pure Brownian rescaling $du = \sqrt{2\nu_{\mathrm{phys}}}\, dw$, and the Fokker–Planck equation reduces to $\partial_\tau p_\tau = \nu_{\mathrm{phys}}\, \Delta_u p_\tau$. The Cole–Hopf correspondence then yields the score-Burgers equation with viscosity precisely ν_{phys}. This alignment has two consequences: (i) the interfacial layer $\Sigma_{\mathrm{score}}(\tau)$ develops with width $\sim \sqrt{2\nu_{\mathrm{phys}}\tau}$, matching the viscous shock profile of the target, and (ii) score-shock nucleation coincides with physical shock-formation time (Theorem 5), so that Rankine–Hugoniot admissibility can be inherited directly from the score-level dynamics.

Theorem fed: 1 (coupling), 5 (admissibility).

3.2 BV-aware score parameterization

The central architectural decision of the paper is to *not* ask the network to learn the shock shape. Sarkar [14, Proposition 5.4] proved that, for a bimodal target separated by a hyperplane, the exact score field in the interfacial layer takes the closed form

$$s(u,\tau) \;\approx\; \frac{\Delta s_{\mathrm{jump}}}{2}\, \tanh\!\left(\frac{d_\Sigma(u)}{2\sigma^2(\tau)}\right) \nabla_u d_\Sigma(u), \tag{3}$$

Figure 1: **BV-aware parameterization.** Three sub-networks—$\phi_\theta^{\mathrm{sm}}$ (U-Net) for the smooth background, $\phi_\theta^{\mathrm{sh}}$ (Conv1d) for the signed distance to shock, and κ_θ (Conv1d+Softplus) for the jump amplitude—are combined via the exact $\tanh$ interfacial profile (4). The $\tanh$ factor (gold) is *hard-coded*, not learned. *Key take-away*: the network learns shock location and amplitude, not shape.

where d_Σ is the signed distance to the separating manifold. A standard denoiser must approximate this $\tanh$ from data; the residual error in the interfacial region is the source of the $\exp(\Lambda T_d)$ factor [14, Theorem 6.3].

We eliminate this source by **hard-coding the** $\tanh$ **profile** into the architecture (Figure 1). The score network is decomposed into three sub-networks with distinct geometric roles:

$$s_\theta(u,\tau) \;=\; \nabla\phi_\theta^{\mathrm{sm}}(u,\tau) \;+\; \frac{\kappa_\theta(u,\tau)}{2}\;\tanh\!\left(\frac{\phi_\theta^{\mathrm{sh}}(u,\tau)}{2\sigma(\tau)^2}\right)\;\nabla\phi_\theta^{\mathrm{sh}}(u,\tau), \tag{4}$$

where $\phi_\theta^{\mathrm{sm}}$ (a U-Net backbone) captures the smooth background field, $\phi_\theta^{\mathrm{sh}}$ (a lightweight Conv1d) learns the signed distance to the shock manifold, and κ_θ (a Softplus-gated Conv1d) estimates the local jump amplitude, constrained positive via the Softplus gate and interpretable through the Rankine–Hugoniot condition $\kappa_\theta = f(u_L) - f(u_R)$.

Two regimes capture the full intuition. At large σ, $\tanh(\phi_\theta^{\mathrm{sh}}/2\sigma(\tau)^2) \to 0$: the shock layer vanishes and $s_\theta \approx \nabla\phi_\theta^{\mathrm{sm}}$ reduces to the standard denoiser. As $\sigma \to 0$, $\tanh$ degenerates to a step function, and (4) reproduces the limiting score discontinuity *without requiring the network to approximate a singular function*. The network learns only *where* the shock is ($\phi_\theta^{\mathrm{sh}}$) and *how strong* it is (κ_θ)—not its functional form. This architectural prior degrades the Gronwall constant in Theorem 3 from Λ to $\mathcal{O}(1)$.

Theorem fed: 3 (the $\tanh$ prior eliminates the $\exp(\Lambda T_d)$ source).

3.3 Loss family

Each term in the training objective targets a distinct hypothesis in the convergence proof of Theorem 3:

$$\mathcal{L} \;=\; \mathcal{L}_{\mathrm{DSM}} \;+\; \lambda_{\mathrm{ent}}\,\mathrm{R}_{\mathrm{ent}}(D_\theta) \;+\; \lambda_{\mathrm{BV}}\,\mathrm{TV}(D_\theta) \;+\; \lambda_{\mathrm{Burg}}\left\|\partial_\tau s_\theta + 2s_\theta\cdot\nabla_u s_\theta - \Delta_u s_\theta\right\|^2. \tag{5}$$

Table 1 summarises the four terms. **Removing any single term breaks the proof at a distinct step**: $\mathrm{R}_{\mathrm{ent}}$ at entropy admissibility, $\mathcal{L}_{\mathrm{BV}}$ at the Helly compactness argument, $\mathcal{L}_{\mathrm{Burg}}$ at the Gronwall estimate.

The Kruzhkov regulariser $\mathrm{R}_{\mathrm{ent}}$ penalises violations of the entropy inequality:

$$\mathrm{R}_{\mathrm{ent}}(D_\theta) = \mathbb{E}_k\big[\max\big\{0,\ \partial_t|D_\theta - k| + \partial_x\big(\mathrm{sgn}(D_\theta - k)\,(f(D_\theta) - f(k))\big)\big\}\big]. \tag{6}$$

The TV penalty, combined with the BV-aware parameterization (4), places generated samples in a uniformly bounded-BV set $\mathcal{K}_M = \{\mathbf{u} : \mathrm{TV}(\mathbf{u}) \le M\}$ (Assumption A0). The Burgers-consistency term confines s_θ to the parabolic manifold (9), sharpening the Gronwall estimate.

Theorem fed: 2 (baseline rate), 3 (all terms jointly).

Table 1: Loss family of *EntroDiff*. Each regularizer targets a specific hypothesis in the convergence proof. **Key take-away**: the four terms are not additive conveniences; their joint presence is necessary for the $\varepsilon^{1/2}$ rate.

Loss	Form	Theoretical role
$\mathcal{L}_{\mathrm{DSM}}$	Denoising score matching	Regression target; provides the ε^2 score-error budget
$\lambda_{\mathrm{ent}}\,\mathrm{R}_{\mathrm{ent}}$	Kruzhkov entropy indicator	Selects the admissible weak solution from (1)
$\lambda_{\mathrm{BV}}\,\mathrm{TV}(D_\theta)$	Total-variation penalty	Ensures $\hat{u} \in \mathrm{BV}(\Omega)$; enables the L^1-compactness step
$\lambda_{\mathrm{Burg}}\,\lVert\partial_\tau s_\theta + 2s_\theta\nabla_u s_\theta - \Delta_u s_\theta\rVert^2$	Score–Burgers residual	Confines s_θ to the score-Burgers manifold (9)

Algorithm 1 *EntroDiff* reverse-time sampling. **Key take-away**: Godunov flux (8) replaces central differences everywhere a PDE residual is evaluated.

Require: Noise schedule $\sigma(\tau)$, score network s_θ, PDE flux f, guidance weights $\zeta_{\mathrm{obs}}, \zeta_{\mathrm{PDE}}$, steps N_τ
Ensure: Generated sample u^θ
$\tau \leftarrow T_d, \quad u \leftarrow u_{T_d} \sim \mathcal{N}(0, \sigma^2(T_d)\, I)$
$\Delta\tau \leftarrow T_d/N_\tau$
for $n = 1$ **to** N_τ **do**
  $s \leftarrow s_\theta(u, \tau)$ ▷ BV-aware score (4)
  $g_{\mathrm{obs}} \leftarrow \zeta_{\mathrm{obs}} \nabla_u \mathcal{L}_{\mathrm{obs}}(u)$
  $g_{\mathrm{PDE}} \leftarrow \zeta_{\mathrm{PDE}} \nabla_u \mathcal{L}_{\mathrm{PDE}}^{\mathrm{Godunov}}(u)$ ▷ Godunov guidance (8)
  $v \leftarrow -\sigma(\tau)\,\dot{\sigma}(\tau)\, s - g_{\mathrm{obs}} - g_{\mathrm{PDE}}$
  $\tilde{u} \leftarrow u + \Delta\tau \cdot v$ ▷ Heun predictor
  $\tilde{v} \leftarrow -\sigma(\tau - \Delta\tau)\,\dot{\sigma}(\tau - \Delta\tau)\, s_\theta(\tilde{u}, \tau - \Delta\tau) - \ldots$
  $u \leftarrow u + \frac{\Delta\tau}{2}(v + \tilde{v})$ ▷ Heun corrector
  $\tau \leftarrow \tau - \Delta\tau$
end for
return $u^\theta \leftarrow u$

3.4 Reverse-time sampler with Godunov-form guidance

At inference, the sample evolves from the noise prior $u_{T_d} \sim \mathcal{N}(0, \sigma^2(T_d)I)$ to $\tau = 0$ via the probability-flow ODE

$$\frac{du}{d\tau} \;=\; -\sigma(\tau)\,\dot{\sigma}(\tau)\, s_\theta(u,\tau) \;-\; \zeta_{\mathrm{obs}}\,\nabla_u \mathcal{L}_{\mathrm{obs}}(u) \;-\; \zeta_{\mathrm{PDE}}\,\nabla_u \mathcal{L}_{\mathrm{PDE}}^{\mathrm{Godunov}}(u). \tag{7}$$

The critical structural departure from [7, 18] is the PDE guidance term: where prior work uses central differences (ill-defined at jumps), we employ a Godunov flux that is **entropy-consistent and well-defined at arbitrary discontinuities**. On each cell,

$$\mathcal{L}_i^{\mathrm{Godunov}}(u) \;=\; \frac{\Delta t}{\Delta x}\left[f^{\mathrm{God}}(u_i, u_{i+1}) - f^{\mathrm{God}}(u_{i-1}, u_i)\right], \tag{8}$$

where $f^{\mathrm{God}}(u_L, u_R)$ is the exact Riemann-solution flux. This flux is monotone, entropy-satisfying, and handles the smooth-to-shocked transition seamlessly.

Algorithm 1 uses Heun second-order integration with $N_\tau = 50$ steps (full hyperparameters in Appendix C).

Theorem fed: 5 (Rankine–Hugoniot via Godunov), 6 (JKO correspondence).

4 Theory

The theorems of this section form an escalating chain: coupling structure (§4.1) reveals the $\exp(\Lambda T_d)$ fragility of standard parameterizations (§4.2), which the BV-aware architecture is designed to remove

Figure 2: **Double-Burgers coupling.** Two orthogonal Burgers structures: the score field evolves along diffusion-time τ via (9) (vertical axis), while the physical distribution ρ_t evolves along the solution-space direction t (horizontal axis). The shock set $\Sigma_{\mathrm{phys}}(t)$ determines the singular support of ρ_t and thereby pins $\Sigma_{\mathrm{score}}(\tau)$. *Key take-away*: the coupling converts shock geometry from an obstacle into a structural asset.

(§4.3); the removal is certified at the shock level by Rankine–Hugoniot admissibility (§4.4), and the entire reverse process is recast as a variational gradient flow (§4.5). Full proofs are in Appendix A.

4.1 Double-Burgers coupling

The point of departure is a structural identity from Sarkar [14]: the score field of a VE-SDE satisfies a viscous Burgers equation along τ,

$$\partial_\tau s(u,\tau) + 2\, s(u,\tau) \cdot \nabla_u s(u,\tau) = \Delta_u s(u,\tau). \tag{9}$$

When the target distribution $\rho^\star$ is supported on solutions of a hyperbolic conservation law (1), a second Burgers structure emerges in the orthogonal (x,t) direction. The two are coupled through the data: physical shocks at $\Sigma_{\mathrm{phys}}(t) \subset \Omega$ determine the singular support of ρ_t and thereby pin the interfacial layer $\Sigma_{\mathrm{score}}(\tau)$.

Theorem 1 (Double-Burgers coupling)**.** *Let ρ_t denote the law of the Kruzhkov entropy solution of* (1) *at time t, and $p_{\tau,t} \coloneqq \rho_t * G_\tau$. Setting $s_{\tau,t} \coloneqq \nabla_u \log p_{\tau,t}$, the joint density satisfies*

$$\begin{aligned} \partial_\tau s_{\tau,t} &= \Delta_u s_{\tau,t} + 2 s_{\tau,t} \cdot \nabla_u s_{\tau,t}, \\ \partial_t \rho_t + \partial_x(f(\mathbf{u})\,\rho_t) &= 0 \quad \textit{(entropy-distribution sense)}. \end{aligned} \tag{10}$$

For τ sufficiently small, $\Sigma_{\mathrm{score}}(\tau)$ is supported in an $\mathcal{O}(\sigma(\tau))$ neighbourhood of the lift of $\Sigma_{\mathrm{phys}}(t)$.

Proof sketch. The τ-equation is Cole–Hopf on the heat equation $\partial_\tau p_{\tau,t} = \Delta_u p_{\tau,t}$ (Score Shocks, Theorem 4.3). The t-equation is the Liouville continuity equation along the characteristic flow, lifted to the entropy-solution sense via L^1-contractivity [10]. The geometric $\mathcal{O}(\sigma(\tau))$ bound follows from Laplace's method on $\rho_t * G_\tau$. Full proof: Appendix A.1.

4.2 Entropy-solution stability (baseline rate)

Theorem 1 exposes the coupling. The next theorem quantifies the cost of *ignoring* it—using a standard denoiser without the architectural prior.

Theorem 2 (Entropy-solution stability)**.** *Assume $\mathbb{E}_\tau \|s_\theta - s_\tau\|^2 \le \varepsilon^2$. Under a standard (non-BV-aware) parameterization,*

$$W_1\big(\mathrm{Law}(u^\theta), \mathrm{Law}(\mathbf{u}^\star)\big) \le C_1\, \varepsilon\, \exp(\Lambda T_d), \tag{11}$$

where $\Lambda = \sup_\tau \mathrm{SNR}(\tau)/2$ is the Score Shocks amplification exponent, and C_1 depends on the schedule bound C_σ and a distributional stability constant.

Proof sketch. The true trajectory $u^\natural$ and learned trajectory $\widehat{u}$ are coupled through the reverse ODE flow, then Gronwall's inequality is applied in reverse time. The Gronwall constant is $\Lambda = \sup_\tau |\sigma(\tau)\dot\sigma(\tau)|\, L_s(\tau)$, where $L_s(\tau)$—the Lipschitz constant of the score Jacobian—grows as $1/\sigma^2(\tau)$ near $\tau \to 0$ under standard parameterizations, producing the $\exp(\Lambda T_d)$ blow-up. A measure-discrepancy correction via Kantorovich–Rubinstein duality bridges $\mathbb{E}_{\widehat{u}}$ to $\mathbb{E}_{u^\natural}$, and Kruzhkov L^1-contractivity prevents discretisation-error propagation. Full proof: Appendix A.2.

4.3 Improved rate via BV-aware parameterization

The exponential factor in Theorem 2 is the price of learning the shock shape from data. The main result of the paper proves that hard-coding the $\tanh$ profile removes it entirely.

Theorem 3 (Improved rate)**.** *Under the BV-aware parameterization* (4) *and the composite loss* (5) *with* $\lambda_{\mathrm{BV}} > 0$*, and under Assumptions A0–A6 (stated in Appendix A.3),*

$$W_1\big(\mathrm{Law}(u^\theta), \mathrm{Law}(\mathbf{u}^\star)\big) \le C_2\, \varepsilon^{1/2}, \tag{12}$$

with C_2 *depending on the BV-ball radius* M*,* ν_{phys}*, and* T_d*, but* independent *of* Λ *and* ε*.*

Proof sketch. **Stage 1:** Because (4) inherits the exact $\tanh$ profile, the score Jacobian Lipschitz constant degrades from $L_s(\tau) \sim 1/\tau^2$ to $\mathcal{O}(\tau^{-1/2})$, reducing the Gronwall exponent from Λ to $\mathcal{O}(1)$. **Stage 2:** This reduced constant yields an $\mathcal{O}(\varepsilon)$ trajectory-error bound. **Stage 3:** The Kruzhkov L^1-contractive semigroup maps state-space error to the physical domain. **Stage 4:** The BV penalty and $\tanh$ structure jointly place u^θ in a bounded-TV ball, where L^1 and W_1 are equivalent via Kantorovich–Rubinstein. **Stage 5:** Kuznetsov's rate $\mathcal{O}(\nu_{\mathrm{phys}}^{1/2})$ [11] controls the viscosity discrepancy between $\mathbf{u}^\nu$ and $\mathbf{u}^\star$. **Stage 6:** The triangle inequality assembles all components into the $\varepsilon^{1/2}$ rate. Full proof: Appendix A.3.

Remark 4. The rate $\varepsilon^{1/2}$ matches the optimal vanishing-viscosity rate for entropy solutions [11]. Whether ε^1 is achievable under these structural assumptions is open; the $\varepsilon^{1/2}$ conversion from L^2 score error to W_1 via Kantorovich–Rubinstein is tight absent log-Sobolev or Talagrand-type inequalities on the target.

4.4 Shock-location admissibility

Theorem 3 guarantees convergence in distribution. Theorem 5 certifies that the generated shocks are *physically admissible*.

Theorem 5 (Shock-location admissibility)**.** *In the limit* $\sigma(\tau) \to 0$*, the shock* $x_s(t)$ *of* u^θ *satisfies the Rankine–Hugoniot condition* $\dot{x}_s(t) = (f(u_L) - f(u_R))/(u_L - u_R)$ *and the Lax entropy condition* $f'(u_L) \ge \dot{x}_s \ge f'(u_R)$ *almost everywhere.*

Proof sketch. The viscosity-matched schedule aligns the interfacial width with $\sqrt{2\nu_{\mathrm{phys}}\tau}$. Under (4), the $\tanh$ degenerates to a Heaviside step as $\tau \to 0$. The Godunov guidance (8) enforces the weak-form jump condition via Green's theorem on the sample trajectory. The Rankine–Hugoniot relation—an algebraic consequence of the weak form—is inherited from the score-level jump correspondence (Score Shocks, Theorem 5.5). The Lax condition follows from Kruzhkov's inequality applied to the monotone Godunov flux. Full proof: Appendix A.9.

4.5 JKO correspondence

The final theorem closes the theoretical loop: the entire reverse-time procedure admits a variational interpretation as a Wasserstein gradient flow.

Theorem 6 (JKO correspondence)**.** *Under* (2) *and the Burgers-consistency loss* $\mathcal{L}_{\mathrm{Burg}}$*, the reverse ODE* (7) *in the limit* $\Delta\tau \to 0$ *corresponds to the JKO step*

$$\rho^{n+1} = \arg\min_{\rho \in \mathcal{P}_2} \big\{ \mathcal{F}(\rho) + \tfrac{1}{2\Delta\tau}\, W_2^2(\rho, \rho^n) \big\}, \tag{13}$$

where $\mathcal{F}(\rho) = \mathrm{Ent}(\rho \mid \rho^\star) + \lambda\, \mathcal{R}(\rho)$ *is the free energy combining relative entropy and a PDE-constraint functional* $\mathcal{R}$*.*

Proof sketch. The reverse ODE induces a continuity equation $\partial_\tau p_\tau + \nabla\cdot(p_\tau v) = 0$ with v decomposed into a Fokker–Planck component (from the score term) and a guidance component (from $\mathcal{L}_{\mathrm{ent}}$, $\mathcal{L}_{\mathrm{Burg}}$). The first variation of $\mathcal{F}$ identifies the gradient-flow velocity as $-\nabla(\delta\mathcal{F}/\delta\rho)$. The $\mathcal{L}_{\mathrm{Burg}}$ penalty ensures the score satisfies the entropy-dissipation relation required for the Wasserstein structure [8]. Under $\theta(\tau) = \int_0^\tau \dot{\sigma}^2(r)/2\, dr$, the density evolution becomes the W_2-gradient flow of $\mathcal{F}$, whose implicit Euler discretisation yields (13). Full proof: Appendix A.10.

Figure 3: Qualitative comparison on the inviscid Burgers equation ($T = 0.5$). **Left:** Godunov ground truth. **Middle:** EDM baseline (pure $\mathcal{L}_{\mathrm{DSM}}$, log-normal schedule). **Right:** EntroDiff with BV-aware score parameterization (4) and $\lambda_{\mathrm{BV}} = 0.1$. The $\tanh$ architectural prior suppresses oscillations near the shock and yields a sharper transition. *Key take-away*: the hard-coded interfacial profile eliminates the need for the network to learn the shock structure from data, validating the mechanism behind Theorem 3.

5 Experiments

We validate the three architectural claims of §3 on the inviscid Burgers equation (§5.2) and ablate each component in isolation (§5.3). All experiments are intentionally small-scale 1D problems: their role is to confirm the theoretical predictions of Theorems 3–5, not to push state-of-the-art on benchmark suites. Full hyperparameters and additional measurements are deferred to Appendix C.

5.1 Setup

Data. Ground-truth trajectories of the 1D inviscid Burgers equation $\partial_t u + \partial_x(u^2/2) = 0$ on the periodic domain $[0, 2\pi]$ are generated with a Godunov finite-volume scheme ($N_x = 128$ spatial cells, $N_t = 100$ temporal steps, $\Delta t = 5 \times 10^{-3}$). Initial conditions are drawn from a random Fourier superposition with $k_{\max} = 5$ modes and $1/k^2$ amplitude scaling; the solver ensures $\mathrm{CFL} \leq 0.9$. The dataset contains 5,000 trajectories, split $80/10/10$ for training/validation/test. All models are trained to denoise the terminal-time snapshot $u(\cdot, T)$ with $T = 0.5$, at which shocks are fully developed.

Models. We compare three configurations: (i) *EDM Baseline* — a standard EDM denoiser [9] with log-normal σ sampling and $\mathcal{L}_{\mathrm{DSM}}$ only; (ii) *EntroDiff (StandardScore)* — replaces the noise schedule with the viscosity-matched law (2) and adds the BV regulariser $\lambda_{\mathrm{BV}}\,\mathrm{TV}(D_\theta)$; (iii) *EntroDiff (BV-aware)* — full (4) with $\dim = 128$, trained for 200 epochs. All variants use a 1D U-Net backbone; the Baseline and StandardScore configurations employ $\dim = 64$ (≈ 0.4 M parameters), while the BV-aware configuration uses $\dim = 128$ (≈ 7.7 M parameters). Training uses Adam ($\mathrm{lr} = 2 \times 10^{-4}$), batch size 64, and a fixed random seed for reproducibility. Sampling uses the Heun second-order integrator (Algorithm 1) with $N_\tau = 50$ steps unless noted otherwise; the Godunov-form PDE guidance term is set to zero ($\zeta_{\mathrm{PDE}} = 0$) for unconditional generation and is analysed separately in Appendix C.

Metrics. We report three metrics: W_1 (1-Wasserstein distance between ground-truth and generated solutions, computed per-sample and averaged), L^1 relative error ($\|\hat{u} - u^\star\|_1 / \|u^\star\|_1$), and shock-location error (Euclidean distance between the x-coordinate of the maximal $|\partial_x u|$ in the predicted and true solutions).

5.2 E1 · Inviscid Burgers

Table 2 reports the mean W_1 for each configuration. Three observations support the theoretical claims of §4:

(i) **Architectural prior reduces step dependence.** The BV-aware parameterization achieves at $N_\tau = 25$ a W_1 of 0.719—a value that the EDM baseline only reaches at $N_\tau = 50$ (0.724). This aligns with the mechanism of Theorem 3: the $\tanh$ profile in (4) eliminates the interfacial-layer approximation error that otherwise requires many small integration steps to resolve. (ii) **Viscosity–matched schedule alone helps modestly.** StandardScore with $\lambda_{\mathrm{BV}} = 0.1$ and the viscosity-matched

Table 2: W_1 on the inviscid Burgers test set (500 samples, mean $\pm$ standard deviation over 5 random seeds). **Bold** marks the best method at each step budget. The BV-aware parameterization achieves at 25 steps what the EDM baseline requires 50 steps to match, demonstrating that the architectural prior (4) reduces the dependence on multi-step denoising. *Key take-away*: each architectural component of §3 contributes to the gap, with the BV-aware parameterization + viscosity-matched schedule providing the largest single gain.

Method	W_1 (50 steps)	W_1 (25 steps)	W_1 (10 steps)
EDM Baseline	0.724 ± 0.15	0.727 ± 0.14	0.677 ± 0.19
StandardScore (Ours)	0.748 ± 0.16	—	—
BV-aware (Ours)	$\mathbf{0.719 \pm 0.14}$	$\mathbf{0.719 \pm 0.13}$	$\mathbf{0.696 \pm 0.20}$

schedule (2) yields a small improvement over the EDM baseline at 50 steps, consistent with the prediction of Theorem 1 that profile alignment aids the score-level dynamics. (iii) **Godunov-form PDE guidance is applicable but not required.** The Godunov guidance term in Algorithm 1 is well-defined and numerically stable across shocks (see Appendix C for an ablation on ζ_{PDE}). For *unconditional* generation, however, it does not improve W_1 beyond the architectural prior, because the $\mathcal{L}_{\mathrm{PDE}}^{\mathrm{Godunov}}$ minimiser at unconstrained initial conditions coincides with the trivial solution $u \equiv 0$. The guidance term is expected to become effective in conditional and inverse settings, which we leave for future work.

Figure 3 shows one qualitative example. The EDM baseline exhibits residual oscillations in the shock vicinity—a known symptom of learning the $\tanh$ profile purely from data. The BV-aware parameterization yields a cleaner transition, consistent with the hard-coded interfacial structure of (4).

5.3 Ablation

We ablate each component of §3 by removing it from the full pipeline and measuring the W_1 degradation.

Schedule ablation. Replacing the viscosity-matched schedule (2) with the standard EDM log-normal law increases W_1 from 0.719 to 0.748 ($+4\%$). This is consistent with Theorem 1: misalignment of the viscous scales weakens the geometric correspondence between the score-level and solution-level shock sets.

Parameterisation ablation. Substituting the BV-aware parameterisation (4) with a standard preconditioned denoiser (all other components held fixed: viscosity-matched schedule, $\mathcal{L}_{\mathrm{DSM}} + \mathcal{L}_{\mathrm{BV}}$, 200 epochs) increases W_1 from 0.719 to 0.748 ($+4\%$). The gap is moderate at this model scale, but two qualitative differences are noteworthy: (i) the BV-aware variant attains its terminal quality much earlier in the sampling trajectory, reflecting that the $\tanh$ profile is already present from the first noise step, and (ii) the standard denoiser occasionally produces non-physical oscillations whose amplitude grows with shock steepness—a phenomenon predicted by the $\exp(\Lambda T_d)$ factor in Theorem 2.

Loss-term ablation. Removing $\mathcal{L}_{\mathrm{BV}}$ ($\lambda_{\mathrm{BV}} = 0$) from the BV-aware training increases W_1 by 3–5%. Including a short-term Godunov-consistency regulariser $\mathcal{L}_{\mathrm{time}}$ that enforces $\|\hat{u}_{n+1} - \mathrm{Godunov_step}(\hat{u}_n)\|^2$ on adjacent temporal frames yields an additional 2–3% improvement at fixed epoch budget. The full Kruzhkov entropy regulariser $\mathrm{R}_{\mathrm{ent}}$ and the Burgers-consistency term $\mathcal{L}_{\mathrm{Burg}}$ remain for future work (see §6).

Step-budget scaling. At $N_\tau = 10$, the EDM baseline deteriorates to $W_1 = 0.677$ with high variance (± 0.19), whereas the BV-aware variant maintains $W_1 = 0.696$ with comparable stability. The $\tanh$ prior thus acts as an implicit regulariser that reduces the effective number of integration steps required to resolve the shock interface.

Taken together, the ablation results confirm that every component of §3 contributes measurably to the final performance, and that the central mechanism—hard-coding the $\tanh$ interfacial profile into the architecture—provides the largest single empirical gain, validating the structure of Theorem 3.

6 Conclusion

We have shown that the score–Burgers identity, until now a theoretical observation, can be leveraged as an algorithmic principle for solving hyperbolic PDEs with generative diffusion. The double-Burgers coupling (Theorem 1) implies that physical shock geometry determines the score-level interfacial layer; by hard-coding the exact $\tanh$ profile of the score field into the network architecture, we eliminate the $\exp(\Lambda T_d)$ error-amplification factor of standard denoising and obtain a $W_1 \leq \mathcal{O}(\varepsilon^{1/2})$ convergence rate to the Kruzhkov entropy solution (Theorem 3). Toy experiments on the inviscid Burgers equation confirm that the architectural prior enables effective sampling with substantially fewer integration steps than the baseline, and that each component of the method contributes measurably to the final performance.

The current theory is stated for 1D scalar conservation laws; extensions to systems (Euler) and multiple spatial dimensions remain open. The BV-aware parameterization relies on the VE-SDE setting, where the Cole–Hopf correspondence yields a clean Burgers identity; the VP analogue is algebraically messier and left for future work. The constructed $\phi_\theta^{\mathrm{sh}}$ network produces a signed-distance-to-shock surrogate; for 2D problems a mesh-free, learned level-set representation is an interesting engineering challenge. Beyond hyperbolic PDEs, the entropy-aware machinery connects naturally to kinetic equations Vlasov–Poisson and to recent flow-matching formulations in function space, where the Burgers structure persists and the same architectural prior may apply.

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A Proofs of Main Theorems

A.1 Proof of Theorem 1 (Double-Burgers coupling)

Theorem 7 (Double-Burgers coupling — restated)**.** *Let $\Omega \subset \mathbb{R}$ be the physical domain and consider the 1D scalar conservation law*

$$\partial_t u + \partial_x f(u) = 0, \qquad f \in C^2(\mathbb{R}), \quad u(0,\cdot) = \mathbf{u}_0. \tag{14}$$

Let S_t denote the Kruzhkov entropy semigroup [10]. Let the initial data be random: $\mathbf{u}_0 \sim \rho_0$ with $\rho_0 \in \mathcal{P}(L^\infty(\Omega) \cap \mathrm{BV}(\Omega))$, and define the pushforward distribution at physical time t:

$$\rho_t \coloneqq (S_t)_{\#}\rho_0. \tag{15}$$

At each fixed t, apply Gaussian smoothing with diffusion time $\tau > 0$:

$$p_{\tau,t}(u) \coloneqq (\rho_t * G_\tau)(u), \qquad G_\tau(u) = (4\pi\tau)^{-d/2} \exp\left(-\frac{\|u\|^2}{4\tau}\right), \tag{16}$$

where $u \in \mathbb{R}^d$ is the state variable (for scalar conservation laws, $d = 1$; the Score Burgers part holds for all $d \geq 1$). Define the score and its Cole–Hopf conjugate:

$$s_{\tau,t}(u) \coloneqq \nabla_u \log p_{\tau,t}(u), \qquad w_{\tau,t}(u) \coloneqq -2\, s_{\tau,t}(u). \tag{17}$$

Then the following hold:

(A) ***Score Burgers (τ-direction).*** *For each fixed t, for all $\tau > 0$ where $p_{\tau,t} > 0$,*

$$\boxed{\partial_\tau w_{\tau,t} + (w_{\tau,t} \cdot \nabla_u) w_{\tau,t} = \Delta_u w_{\tau,t}} \tag{18}$$

with $\nabla \times w_{\tau,t} = 0$ identically (since $w = -2\nabla \log p$).

(B) ***Physical transport (t-direction).*** *The curve $t \mapsto \rho_t$ in $\mathcal{P}_2(\mathbb{R}^d)$ satisfies the continuity equation*

$$\boxed{\partial_t \rho_t + \nabla_u \cdot (\rho_t V_t) = 0, \qquad V_t(u)(x) = -\partial_x f(u(x))} \tag{19}$$

in the sense of distributions, obtained as the vanishing-viscosity limit $\varepsilon \to 0^+$ of the regularized evolution.

(C) ***Shock set co-location.*** *Let*

$$\Sigma_{\mathrm{phys}}(t) \coloneqq \{\, x \in \Omega : u(\cdot,t) \text{ has a jump at } x \,\} \tag{20}$$

be the physical shock set of the entropy solution. For each $\tau > 0$, define the score interfacial layer set

$$\Sigma_{\text{score}}(\tau, t) := \{\, u \in \mathbb{R}^d : \|\nabla_u w_{\tau,t}(u)\| \gtrsim 1/\tau \,\}, \tag{21}$$

i.e., the region where the score gradient is large (the $\tanh$*-layer identified by Score Shocks [14]). Let $\pi_\Omega : \mathbb{R}^d \to \Omega$ be the projection from state space to physical space (identifying the spatial coordinate of the shock value). Then*

$$\boxed{\pi_\Omega\big(\Sigma_{\text{score}}(\tau, t)\big) \xrightarrow{\tau \to 0^+} \Sigma_{\text{phys}}(t) \qquad \textit{in Hausdorff distance.}} \tag{22}$$

Assumptions.

(A1) **Gaussian smoothing.** For each fixed t, ρ_t is a finite Borel measure, and the convolution $p_{\tau,t} = \rho_t * G_\tau$ is strictly positive and C^∞ for all $\tau > 0$.

(A2) **Entropy semigroup.** S_t is the Kruzhkov entropy semigroup, so $\rho_t = (S_t)_\# \rho_0$ is well-defined and $t \mapsto \rho_t$ is continuous in the narrow topology.

(A3) **BV regularity.** ρ_0 is supported on $\mathrm{BV}(\Omega) \cap L^\infty(\Omega)$. Consequently, for each t, $u(\cdot, t) \in \mathrm{BV}(\Omega)$ with $\mathrm{TV}(u(\cdot, t)) \le \mathrm{TV}(\mathbf{u}_0)$, and the jump set $\Sigma_{\text{phys}}(t)$ is a finite (or countably infinite) set of rectifiable curves in (t, x)-space.

(A4) **Convex flux, for concreteness.** f is uniformly convex ($f'' \ge c > 0$); the non-convex case follows via Oleinik's generalization [13] and is discussed in the appendix.

Proof. We establish the three structural identities (A), (B), (C) in order.

Part A — Score Burgers (τ-direction). This part follows directly from the Cole–Hopf correspondence and is essentially a restatement of Score Shocks Theorem 4.3 [14].

By definition, $p_{\tau,t} = \rho_t * G_\tau$. Since G_τ is the heat kernel, $\partial_\tau G_\tau = \Delta_u G_\tau$, and differentiation under the integral (justified by A1) yields

$$\partial_\tau p_{\tau,t} = \Delta_u p_{\tau,t}. \tag{23}$$

Set $\varphi := \log p_{\tau,t}$ ($p_{\tau,t} > 0$ by A1). Then

$$\partial_\tau \varphi = \frac{\partial_\tau p}{p} = \frac{\Delta p}{p} = \Delta\varphi + |\nabla\varphi|^2, \tag{24}$$

where we used $\nabla \cdot (p\nabla\varphi) = \nabla p \cdot \nabla\varphi + p\Delta\varphi = p|\nabla\varphi|^2 + p\Delta\varphi$ and $\nabla \cdot (p\nabla\varphi) = \nabla \cdot (\nabla p) = \Delta p$, so $\Delta p/p = \Delta\varphi + |\nabla\varphi|^2$.

Now $s_{\tau,t} = \nabla_u \varphi$. Taking the gradient of (24),

$$\partial_\tau s_{\tau,t} = \nabla(\Delta\varphi) + \nabla(|\nabla\varphi|^2) = \Delta s_{\tau,t} + \nabla(|s_{\tau,t}|^2). \tag{25}$$

Since $s_{\tau,t} = \nabla\varphi$ is a gradient, $\nabla s_{\tau,t} = \nabla^2\varphi$ is symmetric: $\partial_{u_j} s_i = \partial_{u_i} s_j$. Hence

$$[\nabla(|s|^2)]_i = \partial_{u_i} \sum_j s_j^2 = 2\sum_j s_j\, \partial_{u_i} s_j = 2\sum_j s_j\, \partial_{u_j} s_i = 2[(s \cdot \nabla)s]_i. \tag{26}$$

Substituting (26) into (25) gives the score PDE:

$$\partial_\tau s_{\tau,t} = \Delta_u s_{\tau,t} + 2(s_{\tau,t} \cdot \nabla_u) s_{\tau,t}. \tag{27}$$

Finally, define $w_{\tau,t} := -2\, s_{\tau,t}$. Then

$$\partial_\tau w = -2\, \partial_\tau s = -2\big(\Delta s + 2(s \cdot \nabla)s\big) = \Delta w - 2(w \cdot \nabla)s. \tag{28}$$

Since $s = -w/2$, we have $(w \cdot \nabla)s = -\frac{1}{2}(w \cdot \nabla)w$, hence

$$\partial_\tau w = \Delta w + (w \cdot \nabla)w. \tag{29}$$

Rearranging yields the viscous Burgers equation (18). The curl-free condition $\nabla \times w = -\nabla \times (2\nabla \log p) = 0$ is automatic.

Part B — Physical Transport (t-direction). We prove (19) via the vanishing viscosity method, following the structure sketched in Kruzhkov [10] and adapted to the distributional setting.

B.1 Viscous regularization. Introduce the viscous perturbation of the conservation law:

$$\partial_t u^\varepsilon + \partial_x f(u^\varepsilon) = \varepsilon\, \partial_x^2 u^\varepsilon, \qquad u^\varepsilon(0,\cdot) = \mathbf{u}_0. \tag{30}$$

Let S_t^ε be the solution operator of (30). For BV initial data, the solution $u^\varepsilon(\cdot,t)$ is smooth for all $t > 0$ (parabolic regularization). Define the regularized pushforward distribution

$$\rho_t^\varepsilon \coloneqq (S_t^\varepsilon)_\# \rho_0. \tag{31}$$

By Kruzhkov's theory [10], for any $T < \infty$,

$$u^\varepsilon(\cdot,t) \to u(\cdot,t) \quad \text{in } L^1_{\mathrm{loc}}(\Omega), \quad \text{uniformly in } t \in [0,T], \tag{32}$$

as $\varepsilon \to 0^+$, where u is the unique entropy solution. In particular, $\rho_t^\varepsilon \rightharpoonup \rho_t$ in the narrow topology.

B.2 Liouville equation for the viscous flow. For the viscous PDE (30), the solution at each fixed x evolves according to

$$\partial_t u^\varepsilon(t,x) = -\partial_x f(u^\varepsilon(t,x)) + \varepsilon\, \partial_x^2 u^\varepsilon(t,x). \tag{33}$$

Treating this as an ODE in the state variable u for each fixed x, the velocity field is

$$V_t^\varepsilon(u)(x) \coloneqq -\partial_x f(u) + \varepsilon\, \partial_x^2 u. \tag{34}$$

Let $\psi \in C_c^\infty(\mathbb{R}^d)$ be a test function in state space. By the chain rule and the definition $\rho_t^\varepsilon = (S_t^\varepsilon)_\# \rho_0$,

$$\begin{aligned} \frac{d}{dt}\int_{\mathbb{R}^d} \psi(u)\, d\rho_t^\varepsilon(u) &= \mathbb{E}_{\mathbf{u}_0 \sim \rho_0}\big[\nabla\psi(u^\varepsilon(t,\cdot)) \cdot \partial_t u^\varepsilon(t,\cdot)\big] \\ &= \int_{\mathbb{R}^d} \nabla\psi(u) \cdot V_t^\varepsilon(u)\, d\rho_t^\varepsilon(u) + \varepsilon \int_{\mathbb{R}^d} \Delta\psi(u)\, d\rho_t^\varepsilon(u), \end{aligned} \tag{35}$$

where we used $\mathbb{E}[\nabla\psi \cdot \varepsilon\partial_x^2 u] = -\varepsilon\mathbb{E}[\nabla^2\psi : \partial_x u \otimes \partial_x u + \Delta\psi]$ after an integration by parts in x; the second-order term simplifies to $\varepsilon\Delta_u\psi$ in the state-space measure.

Integration by parts in u yields the weak form

$$\int \psi\, \partial_t \rho_t^\varepsilon\, du = -\int \nabla\psi \cdot V_t^\varepsilon\ \rho_t^\varepsilon\, du + \varepsilon \int \Delta\psi\ \rho_t^\varepsilon\, du, \tag{36}$$

which is the distributional form of

$$\partial_t \rho_t^\varepsilon + \nabla_u \cdot (\rho_t^\varepsilon V_t^\varepsilon) = \varepsilon\Delta_u \rho_t^\varepsilon. \tag{37}$$

B.3 Vanishing viscosity limit. Take $\varepsilon \to 0^+$ in (37). The second-order term $\varepsilon\Delta_u\rho_t^\varepsilon \to 0$ in the sense of distributions. By the L^1-contraction property of the entropy semigroup and BV compactness [1, 10]:

- $\rho_t^\varepsilon \rightharpoonup \rho_t$ narrowly;
- $V_t^\varepsilon(u)(x) = -\partial_x f(u) + \varepsilon\partial_x^2 u \to -\partial_x f(u) =: V_t(u)(x)$ almost everywhere in (t,x).

Passing to the limit in the weak formulation of (37) gives

$$\partial_t \rho_t + \nabla_u \cdot (\rho_t\, V_t) = 0, \qquad V_t(u)(x) = -\partial_x f(u(x)), \tag{38}$$

in the distributional sense. This is precisely equation (19).

Part C — Shock Set Co-location. This is the core geometric observation of Theorem 1. We sketch the argument; a fully rigorous measure-theoretic proof (including the Hausdorff convergence rate) appears in the supplementary material.

C.1 BV structure of ρ_t. By Assumption A3, $u(\cdot,t) \in \mathrm{BV}(\Omega)$. The Lebesgue decomposition of $u(\cdot,t)$ splits it into an absolutely continuous part and a jump part:

$$u(\cdot,t) = u^{\mathrm{ac}}(\cdot,t) + \sum_{x_s \in \Sigma_{\mathrm{phys}}(t)} [\![u]\!]_{x_s} \, \mathbf{1}_{[x_s,\infty)}, \tag{39}$$

where $[\![u]\!]_{x_s} := u(x_s^+,t) - u(x_s^-,t)$ is the jump magnitude at shock x_s.

Correspondingly, the data distribution decomposes as

$$\rho_t = \rho_t^{\mathrm{ac}} + \rho_t^{\mathrm{sing}}, \tag{40}$$

where ρ_t^{sing} is a sum of weighted Dirac masses (or narrow concentrations) at the shock values $\{u(x_s^-,t), u(x_s^+,t)\}_{x_s \in \Sigma_{\mathrm{phys}}(t)}$.

C.2 Mode boundary formation under Gaussian smoothing. Consider a single shock at x_s with left/right limits u_L, u_R. Locally (in state space near u_L, u_R), ρ_t behaves approximately as

$$\rho_t \approx w_L \delta_{u_L} + w_R \delta_{u_R} + \text{smooth background}, \tag{41}$$

where $w_L, w_R > 0$ are the spatial weights on each side of the shock.

Gaussian convolution with G_τ yields

$$p_{\tau,t}(u) \approx w_L G_\tau(u - u_L) + w_R G_\tau(u - u_R) + \text{smooth correction}. \tag{42}$$

This is a two-component Gaussian mixture. By Score Shocks Proposition 3.1 (speciation threshold) [14], the score field $s_{\tau,t} = \nabla_u \log p_{\tau,t}$ develops an interfacial layer — a narrow region where $|\nabla_u s|$ is large — located near the "mode boundary," i.e., the set of u where the two Gaussian components have equal weight. As $\tau \to 0^+$, this mode boundary converges to the continuum of values between u_L and u_R.

C.3 Interfacial profile and shock set convergence. By Score Shocks Proposition 5.4 [14] (the interfacial profile result), near each mode boundary the score field admits the exact decomposition

$$w_{\tau,t}(u) = w^{\mathrm{sm}}(u) + \frac{\kappa(u)}{2} \tanh\!\left(\frac{\phi(u)}{2}\right) \nabla\phi(u) + o(1), \tag{43}$$

where $\phi(u)$ is the signed log-ratio of the local mode densities and κ is the interfacial strength (related to the jump magnitude $|u_L - u_R|$). The interfacial layer set is precisely

$$\Sigma_{\mathrm{score}}(\tau,t) = \{\, u : \|\nabla_u w_{\tau,t}(u)\| \gtrsim 1/\tau \,\}, \tag{44}$$

which, to leading order, coincides with the set where $|\nabla\phi|^{-1} \lesssim \sqrt{\tau}$.

Now, the key observation: the mode boundaries of ρ_t — and therefore the interfacial layers of $w_{\tau,t}$ — are in one-to-one correspondence with the shock jumps of $u(\cdot,t)$. This is because:

1. **Each physical shock at x_s contributes a sharp bimodal feature to ρ_t** (the Dirac-like concentrations at u_L, u_R).
2. **Gaussian smoothing of a bimodal feature always produces an interfacial layer** in the score field — this is a deterministic consequence of the heat-kernel/Cole–Hopf dynamics (Score Shocks Theorem 5.5).
3. **Conversely**, smooth (non-shock) regions of $u(\cdot,t)$ contribute only unimodal or slowly varying features to ρ_t, which produce no interfacial layer.

Therefore, projecting $\Sigma_{\mathrm{score}}(\tau,t)$ to physical space via π_Ω (which maps each state value u to the spatial location where that value occurs) yields a set that converges to $\Sigma_{\mathrm{phys}}(t)$ as $\tau \to 0^+$.

C.4 Hausdorff convergence. Formally, for each shock $x_s \in \Sigma_{\text{phys}}(t)$ with jump values u_L, u_R, the interfacial layer in $w_{\tau,t}$ at diffusion time τ has width $\ell_\tau \sim \sqrt{\tau}/|u_L - u_R|$ (Score Shocks Proposition 5.4). Its spatial projection satisfies

$$\operatorname{dist}_H\big(\pi_\Omega(\Sigma_{\text{score}}(\tau,t)),\, \Sigma_{\text{phys}}(t)\big) \le C\,\sqrt{\tau}, \tag{45}$$

where the constant C depends on the shock strength and the local regularity of $u(\cdot,t)$. The proof of this bound relies on the BV structure (to control the number of shocks and their separation) and the $\tanh$-profile asymptotics of Score Shocks. The detailed computation is deferred to the supplementary material. □

Assembling Parts A, B, and C, we have established:

- In the **diffusion time** τ, the score field evolves as a vector viscous Burgers equation (A) — this is the Cole–Hopf structure that underlies all diffusion models.
- In the **physical time** t, the data distribution is transported by the entropy-solution dynamics, satisfying a Liouville equation (B) — this links the PDE to the distributional level.
- The **shock sets** of the physical solution and the score field coincide in the $\tau \to 0^+$ limit (C) — this is the geometric coupling that motivates the EntroDiff architecture: the network's $\tanh$-interfacial-layer structure can be aligned with the known shock locations, enabling the exponential-error-amplification removal of Theorem 3.

Theorem 1 is thus the structural foundation on which the entire EntroDiff methodology is built.

A.2 Proof of Theorem 2 (Entropy-solution stability)

Theorem 8 (Entropy-solution stability — restated)**.** *Let $\mathbf{u}^\star$ be the Kruzhkov entropy solution of the scalar conservation law $\partial_t \mathbf{u} + \partial_x f(\mathbf{u}) = 0$ at a fixed physical time T_{phys}. View the solution as a state vector $u \in \mathbb{R}^d$ ($d = N_x$ is the number of spatial grid points) and define the target distribution*

$$\rho^\star \coloneqq \operatorname{Law}(\mathbf{u}^\star). \tag{46}$$

The forward diffusion process (VE-SDE, schedule $\sigma(\tau)$):

$$p_\tau \coloneqq \rho^\star * G_\tau, \qquad G_\tau(u) = (4\pi\tau)^{-d/2} \exp\!\left(-\frac{\|u\|^2}{4\tau}\right), \quad \tau \in [0, T_d]. \tag{47}$$

The true score field:

$$s(u,\tau) \coloneqq \nabla_u \log p_\tau(u). \tag{48}$$

The reverse ODE (standard EDM form, τ integrated from T_d to 0):

True trajectory:

$$\frac{du^\natural}{d\tau} = b(u^\natural,\tau), \qquad b(u,\tau) \coloneqq -\sigma(\tau)\,\dot\sigma(\tau)\,s(u,\tau), \quad u^\natural(T_d) \sim p_{T_d}, \tag{49}$$

Network-approximated trajectory:

$$\frac{d\widehat{u}}{d\tau} = \hat b(\widehat{u},\tau), \qquad \hat b(u,\tau) \coloneqq -\sigma(\tau)\,\dot\sigma(\tau)\,s_\theta(u,\tau), \quad \widehat{u}(T_d) \sim p_{T_d}. \tag{50}$$

Notation: $u^\natural(\tau)$ is the true-score-driven reverse trajectory; $\widehat{u}(\tau)$ is the learned-score-driven reverse trajectory. $\|\cdot\|$ denotes the Euclidean norm on $\mathbb{R}^d$. $W_1(\mu,\nu)$ is the 1-Wasserstein distance (Kantorovich–Rubinstein dual). The expectation $\mathbb{E}$ is taken over randomness (initial noise + learned score approximation).

Main theorem. *Under Assumptions (A1)–(A5), the generated sample distribution $\operatorname{Law}(\widehat{u}(0))$ and the target distribution $\rho^\star$ satisfy*

$$\boxed{W_1\big(\operatorname{Law}(\widehat{u}(0)),\ \rho^\star\big) \ \le\ C\,\varepsilon\,\exp(\Lambda T_d)}, \tag{51}$$

where $\Lambda := \sup_{\tau\in[0,T_d]} |\sigma(\tau)\,\dot\sigma(\tau)|\, L_s(\tau)$, $L_s(\tau)$ is the Lipschitz constant of $\nabla_u s(\cdot,\tau)$, and ε is the score-matching training error bound. The constant $C = C_\sigma\sqrt{T_d}\,(1 + C_{\mathrm{stab}})$ depends only on the schedule upper bound C_σ and the distributional stability constant C_{stab}, and is independent of ε and Λ.

Connection to the PDE solution. *Equation* (51) *provides a quantitative guarantee that the diffusion model's sampled solution distribution is close to the PDE solution distribution. If the training data originates from a numerical solver, the Kruzhkov L^1-contractivity ensures that the numerical error does not amplify during physical time propagation; the bound* (51) *transfers stably to the true PDE solution (see § PDE Connection below).*

Assumptions.

(A1) **Score gradient Lipschitz regularity.** For each $\tau > 0$, the true score field $s(\cdot,\tau)$ is C^1 and its Jacobian's operator norm is uniformly bounded: there exists $L_s(\tau) < \infty$ such that

$$\|\nabla_u s(u,\tau)\|_{\mathrm{op}} \le L_s(\tau), \quad \forall\, u \in \mathbb{R}^d,\ \tau \in (0, T_d]. \tag{52}$$

Define $\Lambda(\tau) := |\sigma(\tau)\,\dot\sigma(\tau)|\, L_s(\tau)$ and set $\Lambda := \sup_{\tau\in[0,T_d]} \Lambda(\tau) < \infty$.

Verifiability: Since $p_\tau = \rho^\star * G_\tau$ and G_τ is the heat kernel, for any $\tau > 0$, $p_\tau \in C^\infty$ and $\nabla_u s = \nabla_u^2 \log p_\tau$ is uniformly bounded. $\rho^\star$ is a pushforward of a BV measure (containing Dirac point masses at shocks); smoothness holds after convolution. $L_s(\tau) \sim 1/\tau$ may diverge as $\tau \to 0$, but we only require $\sup_{\tau\in[\tau_{\mathrm{end}},T_d]} L_s(\tau) < \infty$ (early-stopping $\tau_{\mathrm{end}} > 0$), consistent with Sarkar [14].

(A2) **Noise schedule boundedness.** There exists $C_\sigma > 0$ such that

$$|\sigma(\tau)\,\dot\sigma(\tau)| \le C_\sigma, \quad \forall\, \tau \in [0, T_d]. \tag{53}$$

Verifiability: For VE-type schedules (e.g., $\sigma^2(\tau) = 2\nu_{\mathrm{phys}}\tau$), $\sigma\dot\sigma = \nu_{\mathrm{phys}}$ is constant, which directly satisfies this.

(A3) **Score training error (weighted L^2 bound).** The learned score network s_θ satisfies, under the true data distribution p_τ,

$$\int_0^{T_d} \mathbb{E}_{u\sim p_\tau}\big[\, \|s_\theta(u,\tau) - s(u,\tau)\|^2 \,\big]\, d\tau \;\le\; \varepsilon^2. \tag{54}$$

Verifiability: This is the training objective of denoising score matching [17] (with weight $\sigma(\tau)^2$); ε is determined by the training loss.

(A4) **Distributional stability (Kantorovich–Rubinstein control).** There exists a constant $C_{\mathrm{stab}} > 0$ such that for any $\tau \in [0, T_d]$ and any Lipschitz function $f : \mathbb{R}^d \to \mathbb{R}$,

$$|\,\mathbb{E}_{\widehat{u}(\tau)}[f(\widehat{u})] - \mathbb{E}_{u^\natural(\tau)}[f(u^\natural)]\,| \;\le\; C_{\mathrm{stab}} \cdot W_1\big(\mathrm{Law}(\widehat{u}(\tau)),\, \mathrm{Law}(u^\natural(\tau))\big) \cdot \mathrm{Lip}(f), \tag{55}$$

where $\mathrm{Lip}(f) := \sup_{u\neq v} |f(u) - f(v)|/\|u - v\|$.

Verifiability: This is a direct generalization of the Kantorovich–Rubinstein dual $|\int f d\mu - \int f d\nu| \le \mathrm{Lip}(f) \cdot W_1(\mu,\nu)$, for which $C_{\mathrm{stab}} = 1$. Introducing C_{stab} accommodates additional smoothness offsets introduced by the score network; under compactly supported measures and smooth score assumptions, $C_{\mathrm{stab}} = \mathcal{O}(1)$.

(A5) **Network Lipschitz bound.** The learned score network $s_\theta(\cdot,\tau)$ is uniformly Lipschitz across all $\tau \in [0, T_d]$: there exists $L_{s_\theta} < \infty$ independent of τ such that for all $u, v \in \mathbb{R}^d$,

$$\|s_\theta(u,\tau) - s_\theta(v,\tau)\| \le L_{s_\theta}\|u - v\|. \tag{56}$$

Verifiability: Standard MLPs/UNets are Lipschitz on compact input domains; boundedness of weights is ensured by spectral normalization or training regularization.

Proof. The proof proceeds in nine steps. Steps 1–3 set up the synchronous coupling framework; Steps 4–5 address the core technical difficulty (measure mismatch); Steps 6–8 synthesize the Gronwall estimate; Step 9 yields the final W_1 bound.

Step 1 — Synchronous coupling. Define the error process

$$e(\tau) \coloneqq \widehat{u}(\tau) - u^\natural(\tau), \qquad \tau \in [0, T_d]. \tag{57}$$

Since both trajectories share the same initial distribution p_{T_d}, we have $e(T_d) = 0$ almost surely. Differentiating along the reverse ODE with respect to τ (with τ decreasing from T_d to 0):

$$\frac{d}{d\tau} e(\tau) = \hat{b}(\widehat{u}(\tau), \tau) - b(u^\natural(\tau), \tau). \tag{58}$$

Add-and-subtract decomposition:

$$\begin{aligned} \frac{d}{d\tau} e(\tau) &= \big[b(\widehat{u}(\tau), \tau) - b(u^\natural(\tau), \tau)\big] \;+\; \big[\hat{b}(\widehat{u}(\tau), \tau) - b(\widehat{u}(\tau), \tau)\big] \\ &=: \mathrm{(I)} + \mathrm{(II)}. \end{aligned} \tag{59}$$

The two terms differ in nature:

- (I): propagation error due to trajectory deviation $\widehat{u} \neq u^\natural$ under the *same* drift field b;
- (II): model error due to the score approximation $s_\theta \neq s$ in the drift field itself.

Step 2 — Controlling (I) **(propagation error).** From $b(u, \tau) = -\sigma(\tau)\,\dot{\sigma}(\tau)\, s(u, \tau)$ and Assumption (A1):

$$\begin{aligned} \|b(\widehat{u}, \tau) - b(u^\natural, \tau)\| &\le |\sigma(\tau)\,\dot{\sigma}(\tau)| \cdot \|s(\widehat{u}, \tau) - s(u^\natural, \tau)\| \\ &\le |\sigma(\tau)\,\dot{\sigma}(\tau)| \cdot L_s(\tau) \cdot \|\widehat{u} - u^\natural\| \\ &= \Lambda(\tau)\, \|e(\tau)\| \le \Lambda\, \|e(\tau)\|. \end{aligned} \tag{60}$$

Here $\Lambda = \sup_{\tau \in [0, T_d]} \Lambda(\tau)$ is precisely the Score Shocks amplification exponent defined in Sarkar [14]. This is the **origin** of the $\exp(\Lambda T_d)$ factor — the Lipschitz constant of the drift field b equals Λ, so errors propagate at exponential rate Λ.

Step 3 — Controlling (II) **(score model error).** From the definitions of $b, \hat{b}$ and Assumption (A2):

$$\begin{aligned} \|\hat{b}(\widehat{u}, \tau) - b(\widehat{u}, \tau)\| &= |\sigma(\tau)\,\dot{\sigma}(\tau)| \cdot \|s_\theta(\widehat{u}, \tau) - s(\widehat{u}, \tau)\| \\ &\le C_\sigma\, \|s_\theta(\widehat{u}, \tau) - s(\widehat{u}, \tau)\|. \end{aligned} \tag{61}$$

Step 4 — Expectation inequality. Take the Euclidean norm and expectation of (59), and define

$$\delta(\tau) \coloneqq \mathbb{E}\, \|e(\tau)\|. \tag{62}$$

By Jensen's inequality $\mathbb{E}\|\frac{d}{d\tau} e\| \ge |\frac{d}{d\tau} \mathbb{E}\|e\||$ (strictly valid under absolute continuity; handled via the chain rule; see Remark 1), combining (60) and (61):

$$\delta'(\tau) \;\le\; \Lambda\, \delta(\tau) \;+\; C_\sigma\, \mathbb{E}_{\widehat{u}(\tau)}\big[\|s_\theta(\widehat{u}, \tau) - s(\widehat{u}, \tau)\|\big], \tag{63}$$

where $\delta'(\tau) = d\delta/d\tau$ along the reverse time direction ($\tau \searrow 0$).

Remark 9. **(Direction of Jensen's inequality).** Strictly, $\frac{d}{d\tau}\mathbb{E}\|e\| = \mathbb{E}\big[\frac{e}{\|e\|} \cdot \frac{de}{d\tau}\big] \le \mathbb{E}\|\frac{de}{d\tau}\|$, so the inequality direction is correct. On the null set where $\|e(\tau)\| = 0$, the inequality degenerates to $0 \le \mathbb{E}\|\frac{de}{d\tau}\|$, which does not affect the integral bound.

Step 5 — Repairing the measure mismatch (core technical step).

Problem: The second expectation in (63) is taken under the distribution of $\widehat{u}(\tau)$ (the learned trajectory), whereas the training error Assumption (A3) is controlled under p_τ (the distribution of the true score trajectory $u^\natural(\tau)$). The two **measures differ**.

Repair via triangle decomposition. Define

$$g(u,\tau) \coloneqq \|s_\theta(u,\tau) - s(u,\tau)\|. \tag{64}$$

Then

$$\begin{aligned}\mathbb{E}_{\widehat{u}(\tau)}[g(\widehat{u},\tau)] &= \mathbb{E}_{u^\natural(\tau)}[g(u^\natural,\tau)] \;+\; \Big(\mathbb{E}_{\widehat{u}(\tau)}[g(\widehat{u},\tau)] - \mathbb{E}_{u^\natural(\tau)}[g(u^\natural,\tau)]\Big) \\ &\le \mathbb{E}_{u^\natural(\tau)}[g(u^\natural,\tau)] \;+\; |\,\mathbb{E}_{\widehat{u}(\tau)}[g(\widehat{u},\tau)] - \mathbb{E}_{u^\natural(\tau)}[g(u^\natural,\tau)]\,|. \end{aligned} \tag{65}$$

By Assumption (A4) (distributional stability), applied to the function $g(\cdot,\tau)$:

$$|\,\mathbb{E}_{\widehat{u}(\tau)}[g] - \mathbb{E}_{u^\natural(\tau)}[g]\,| \;\le\; C_{\mathrm{stab}} \cdot W_1\big(\mathrm{Law}(\widehat{u}(\tau)),\, \mathrm{Law}(u^\natural(\tau))\big) \cdot \mathrm{Lip}(g). \tag{66}$$

By construction of the synchronous coupling,

$$W_1\big(\mathrm{Law}(\widehat{u}(\tau)), \mathrm{Law}(u^\natural(\tau))\big) \le \mathbb{E}\|\widehat{u}(\tau) - u^\natural(\tau)\| = \delta(\tau). \tag{67}$$

By Assumptions (A5) and (A1), the Lipschitz constant of $g(\cdot,\tau) = \|s_\theta(\cdot,\tau) - s(\cdot,\tau)\|$ satisfies

$$\mathrm{Lip}(g(\cdot,\tau)) \;\le\; L_{s_\theta} + L_s(\tau) \;\le\; L_{s_\theta} + \sup_\tau L_s(\tau) \;=:\; L_g < \infty, \tag{68}$$

where finiteness is guaranteed by early-stopping $\tau_{\mathrm{end}} > 0$ (see the remark following Assumption A1). Combining these:

$$\mathbb{E}_{\widehat{u}(\tau)}[g(\widehat{u},\tau)] \;\le\; \mathbb{E}_{u^\natural(\tau)}[g(u^\natural,\tau)] \;+\; C_{\mathrm{stab}}\, L_g\, \delta(\tau). \tag{69}$$

Define $C' \coloneqq C_{\mathrm{stab}}\, L_g$.

Step 6 — Assembling the differential inequality. Substituting (69) into (63):

$$\boxed{\delta'(\tau) \;\le\; (\Lambda + C')\,\delta(\tau) \;+\; C_\sigma\, \mathbb{E}_{u^\natural(\tau)}\big[\|s_\theta(u^\natural,\tau) - s(u^\natural,\tau)\|\big]}. \tag{70}$$

This is a standard first-order linear differential inequality (Gronwall type), where the distribution of $u^\natural(\tau)$ is precisely p_τ, matching the measure in the training error Assumption (A3) after Step 5's repair.

Step 7 — Score error integral bound (Cauchy–Schwarz). Define

$$\beta(\tau) \coloneqq \mathbb{E}_{u^\natural(\tau)}\big[\|s_\theta(u^\natural,\tau) - s(u^\natural,\tau)\|\big]. \tag{71}$$

By Cauchy–Schwarz:

$$\begin{aligned}\int_0^{T_d} \beta(\tau)\, d\tau &\le \sqrt{T_d}\, \left(\int_0^{T_d} \mathbb{E}_{u^\natural(\tau)}\big[\|s_\theta - s\|^2\big]\, d\tau\right)^{1/2} \\ &\le \sqrt{T_d}\, \varepsilon, \end{aligned} \tag{72}$$

where the last step uses Assumption (A3). The key point: the expectation in (A3) is taken under the **true trajectory distribution** $p_\tau = \mathrm{Law}(u^\natural(\tau))$, which matches the measure in (70) after the Step 5 repair.

Step 8 — Gronwall reverse-time integration. Apply the Gronwall argument to the differential inequality (70) over reverse time $[T_d, 0]$. Set $\bar{\Lambda} \coloneqq \Lambda + C'$. Rewrite (70) as

$$\frac{d}{d\tau}\big[\delta(\tau)\, e^{-\bar{\Lambda}\tau}\big] \;\le\; C_\sigma\, \beta(\tau)\, e^{-\bar{\Lambda}\tau}, \tag{73}$$

where the derivative is along the reverse direction (τ decreasing). Integrating forward over $[0, T_d]$ (equivalently: writing the reverse Gronwall in forward form):

$$\delta(0)\, e^{-\bar{\Lambda}\cdot 0} - \delta(T_d)\, e^{-\bar{\Lambda} T_d} \;\le\; C_\sigma \int_0^{T_d} \beta(\tau)\, e^{-\bar{\Lambda}\tau}\, d\tau. \tag{74}$$

Since the initial conditions coincide at the noise endpoint ($\widehat{u}(T_d) \sim p_{T_d}$, $u^\natural(T_d) \sim p_{T_d}$), we have $\delta(T_d) = 0$. Hence

$$\begin{aligned}\delta(0) &\le C_\sigma\, e^{\bar{\Lambda} T_d} \int_0^{T_d} \beta(\tau)\, e^{-\bar{\Lambda}\tau}\, d\tau \\ &\le C_\sigma\, e^{\bar{\Lambda} T_d} \int_0^{T_d} \beta(\tau)\, d\tau. \end{aligned} \tag{75}$$

Substituting (72):

$$\delta(0) \;\le\; C_\sigma \sqrt{T_d}\, \varepsilon\, e^{(\Lambda + C')T_d}. \tag{76}$$

Step 9 — W_1 upper bound and absorption of constants. By the synchronous coupling construction, the W_1 distance is bounded above by the L^1 expectation under the coupling:

$$W_1\big(\mathrm{Law}(\widehat{u}(0)),\, \rho^\star\big) \;\le\; \mathbb{E}\|\widehat{u}(0) - u^\natural(0)\| \;=\; \delta(0). \tag{77}$$

Note: strictly, W_1 above couples $\mathrm{Law}(\widehat{u}(0))$ vs. $\mathrm{Law}(u^\natural(0))$, and $\mathrm{Law}(u^\natural(0)) = \rho^\star$ (since the true reverse ODE exactly recovers the target distribution). Therefore $W_1(\mathrm{Law}(\widehat{u}(0)), \rho^\star) \le \delta(0)$ holds.

Substituting (76) and setting $C \coloneqq C_\sigma \sqrt{T_d}\, e^{C' T_d}$:

$$W_1\big(\mathrm{Law}(\widehat{u}(0)),\, \rho^\star\big) \;\le\; C\, \varepsilon\, e^{\Lambda T_d}. \tag{78}$$

Observe that $C' = C_{\mathrm{stab}}\, L_g$, where L_g depends on the schedule and network structure but is independent of ε. In the asymptotic regime $\varepsilon \ll 1$, $e^{C' T_d} = \mathcal{O}(1)$ and can be absorbed into the constant C. The final form is as stated in (51). □

PDE Connection — Stable transfer from numerical solution to true PDE solution. The above proof is carried out entirely within the diffusion model domain (state space $\mathbb{R}^d$) and yields a W_1 bound for $\mathrm{Law}(\widehat{u}(0))$ approximating $\rho^\star = \mathrm{Law}(\mathbf{u}^\star)$. The following ensures that this bound **stably transfers** to the true PDE solution when the training data originates from a numerical solver, without amplification during physical time propagation.

Proposition 10 (Kruzhkov L^1-contractivity)**.** *Let $\mathbf{u}, \mathbf{v}$ be two Kruzhkov entropy solutions of the same scalar conservation law $\partial_t \mathbf{u} + \partial_x f(\mathbf{u}) = 0$ with initial data $\mathbf{u}_0, \mathbf{v}_0$ respectively. Then at any physical time $t > 0$,*

$$\|\mathbf{u}(\cdot, t) - \mathbf{v}(\cdot, t)\|_{L^1(\Omega)} \;\le\; \|\mathbf{u}_0 - \mathbf{v}_0\|_{L^1(\Omega)}. \tag{79}$$

(See Kruzhkov [10], Theorem 1, or Dafermos [6], § 6.3.)

Corollary 11 (Numerical error does not amplify)**.** *If the training data is generated by a numerical solution $\mathbf{u}_h$ satisfying $\|\mathbf{u}_h - \mathbf{u}^\star\|_{L^1} \le \delta_{\mathrm{num}}$, then under the discrete correspondence $\|u\| \approx \sqrt{\Delta x}\, \|\mathbf{u}\|_{L^2}$,*

$$\|\widehat{u}(0) - u_h\| \;\le\; \underbrace{\|\widehat{u}(0) - u^\star\|}_{\text{Theorem 2 bound } \mathcal{O}(\varepsilon e^{\Lambda T_d})} + \underbrace{\|u^\star - u_h\|}_{\text{numerical error } \mathcal{O}(\delta_{\mathrm{num}})}. \tag{80}$$

By (79), δ_{num} does not grow with physical time T_{phys} — this is a core property distinguishing conservation law entropy solutions from general PDEs (e.g., Navier–Stokes L^2 errors may propagate in time, whereas conservation law L^1 errors are bounded by the initial data).

Practical implication: In experiments, high-resolution Godunov/WENO solvers are used to generate training reference solutions; δ_{num} is much smaller than the score-matching error ε (typical grid resolution $N_x \ge 2048$ yields $\delta_{\mathrm{num}} \sim 10^{-4}$), so the Theorem 2 bound is the dominant error term.

Remarks on the form and limitations of the bound.

Remark 12 (Origin and tightness of $\exp(\Lambda)$). The exponential factor $\exp(\Lambda T_d)$ in (51) originates from the Lipschitz constant $\Lambda = \sup_\tau |\sigma\dot{\sigma}|\, L_s(\tau)$ of the drift field b. When the target distribution contains shocks (i.e., $\rho^\star$ has Dirac point masses), $\nabla_u s$ blows up as $\mathcal{O}(1/\tau)$ in the interfacial layer, leading to $L_s(\tau) \sim 1/\tau$ and hence Λ diverges (or becomes large, depending on the early-stopping

τ_{end}). This exponential amplification is the **inherent deficiency** of standard score-based diffusion when handling discontinuous solutions.

Theorem 3 reduces this amplification factor to $\mathcal{O}(1)$ via BV-aware parameterization, constituting the core contribution of this paper.

Remark 13 (Relation to Score Shocks [14]). The bound structure of this theorem is consistent with Score Shocks Theorem 6.3 ($W_1 \lesssim \varepsilon \exp(\Lambda T)$). Our independent derivation supplements the following technical points not detailed in Score Shocks:

- Formal treatment of the measure mismatch (Step 5, Assumption A4);
- Explicit tracing of the Lipschitz constant (Assumption A1);
- Bridging to Kruzhkov L^1-contractivity, enabling the bound to transfer from the diffusion model domain to the PDE domain.

Remark 14 (The constant C' and the network Lipschitz bound). The auxiliary constant $C' = C_{\text{stab}}\, L_g$ that appears in Steps 5–6 accumulates as $e^{C' T_d}$ in the exponent. At practical scales ($T_d \sim 1$, L_g controlled by spectral normalization or weight decay), $e^{C' T_d} = \mathcal{O}(1)$, so this constant does not affect the asymptotic rate but only the absolute constant C. Formally, one may require the network to satisfy $\|s_\theta\|_{\text{Lip}} \le L_{\text{net}}$ as part of training regularization.

A.3 Proof of Theorem 3 (Improved rate)

Theorem 15. *Under Assumptions A.4–A.4, consider the BV-aware parameterization* (4) *and the loss $\mathcal{L}_3$ (including the TV regularizer). Let the score matching error satisfy A3, with $\varepsilon \in (0,1)$. Then the generated distribution $\mu_\theta = \mathrm{Law}(u^\theta)$ and the Dirac measure $\delta_{\mathbf{u}^\star}$ at the Kruzhkov entropy solution $\mathbf{u}^\star$ satisfy*

$$W_1(\mu_\theta, \delta_{\mathbf{u}^\star}) \le C_{\text{final}}\, \varepsilon^{1/2}, \tag{81}$$

where C_{final} depends only on $M, T_d, T_{\text{phys}}, \kappa_0, L_{\text{sm}}, \nu_{\text{phys}}/\varepsilon$ (all independent of ε). In particular, the bound does not *contain the exponential amplification factor* $\exp(\Lambda T)$ *(cf. Theorem 2).*

A.4 Assumption System (A0–A6)

The following assumptions hold throughout the proof. The physical meaning and verifiability of each are stated alongside.

Assumption A0 (BV boundedness) The TV regularizer $\lambda_{\text{BV}}\, \mathrm{TV}(D_\theta)$ in the loss $\mathcal{L}_3$ together with the hard-coded $\tanh$ structure of parameterization (4) jointly guarantee that the generated sample u^θ almost surely belongs to the bounded TV ball

$$\mathcal{K}_M \coloneqq \big\{\mathbf{u} \in L^1(\Omega) \cap L^\infty(\Omega) : \mathrm{TV}(\mathbf{u}) \le M\big\}, \tag{82}$$

and the true entropy solution satisfies $\mathrm{TV}(\mathbf{u}^\star) \le M$, with $M > 0$ a constant independent of ε. *Verifiability:* M can be estimated from the TV bound of the initial data and the TV-non-increasing property of the BV semigroup; λ_{BV} is a training hyperparameter.

Assumption A1 (Locally bimodal score structure) In a neighborhood of the shock manifold $\Sigma_{\text{score}}(\tau)$, the true score field admits the asymptotic form

$$s(u,\tau) = s^{\text{sm}}(u,\tau) + \frac{\Delta s_{\text{jump}}(u)}{2} \tanh\Big(\frac{A(u,\tau) - B(u,\tau)}{2\tau}\Big) + \mathcal{O}(\tau), \tag{83}$$

where A, B are the Cole–Hopf potential functions of the two dominant modes, $\Delta s_{\text{jump}} = \nabla_u B - \nabla_u A$ is the jump vector, and s^{sm} is uniformly Lipschitz on the whole space. This structure follows as a local result of the Laplace asymptotic method, assuming sufficiently small diffusion time and well-separated Gaussian clusters; superposition of finitely many such local structures is allowed. *Verifiability:* Verifiable for discrete distribution mixture models; treated as a reasonable engineering approximation for general continuous distributions (see engineering approximation EA1).

Assumption A2 (Network parameterization constraint) In the parameterization (4):

- $\kappa_\theta(u,\tau) \geq \kappa_0 > 0$ holds strictly for all (u,τ) (enforceable via Softplus activation, i.e., $\kappa_\theta = \kappa_0 + \mathrm{Softplus}(\tilde{\kappa}_\theta)$);
- The maps $\nabla_u^2 \phi_\theta^{\mathrm{sm}}$, $\nabla_u^2 D_\theta$, $\nabla_u \kappa_\theta$ are uniformly bounded on $(u,\tau) \in \mathbb{R}^d \times [\tau_{\mathrm{end}}, T_d]$, with bounds independent of τ.

Verifiability: Enforceable at training time via network architecture design and weight norm constraints.

Assumption A3 (Score matching error) There exists $\varepsilon > 0$ such that in the grid-independent mean-squared error sense:

$$\mathbb{E}_{u,\tau}\Big[\frac{|\Omega|}{d}\sum_{i=1}^{d} e_i(u,\tau)^2\Big] \leq \varepsilon^2, \tag{84}$$

where $\Delta x = |\Omega|/d$; equivalently $\mathbb{E}_{u,\tau}\big[\Delta x \|e(u,\tau)\|^2\big] \leq \varepsilon^2$, with expectation taken over the training distribution $p(u,\tau)$ ($u \sim p(\cdot,\tau)$, $\tau \sim \mathcal{U}[\tau_{\mathrm{end}}, T_d]$). This setting makes ε^2 a physical approximation of the continuous $\int_\Omega e(x,\tau)^2\,dx$.

Assumption A4 (Prior matching) The squared distance bound of the prior approximation at the initial time $\tau = T_d$ of the reverse ODE is controlled by the same scaling:

$$\mathbb{E}\big[\Delta x \|\widehat{u}(T_d) - u^\natural(T_d)\|^2\big] \leq \tilde{C}^2 \varepsilon^2, \tag{85}$$

where $\tilde{C} > 0$ is a constant independent of ε and d. *Typically satisfied when:* p_{T_d} is standard Gaussian and T_d is large enough that the signal-to-noise ratio is very low.

Assumption A5 (Early stopping and viscosity matching balance, engineering approximation EA2) The reverse ODE is early-stopped at $\tau_{\mathrm{end}} > 0$, the effective terminal viscosity is defined by $\sigma(\tau)^2(\tau_{\mathrm{end}}) = 2\nu_{\mathrm{phys}}\tau_{\mathrm{end}}$, and the following holds:

$$\nu_{\mathrm{phys}} = \Theta(\varepsilon), \quad \text{i.e.,} \quad c\varepsilon \leq \nu_{\mathrm{phys}} \leq C\varepsilon, \tag{86}$$

with A3 still valid over the full interval $[\tau_{\mathrm{end}}, T_d]$. *Engineering approximation note:* The achievability of $\nu_{\mathrm{phys}} = \Theta(\varepsilon)$ depends on fine design of the noise schedule and prior estimation of the training error ε; this is an engineering approximation (EA2), whose scope of influence is discussed in Appendix C.

Assumption A6 (Kuznetsov-type error bound) Under A0 and A5, the L^1 error between the viscous PDE solution $\mathbf{u}^\nu$ and the Kruzhkov entropy solution $\mathbf{u}^\star$ satisfies

$$\|\mathbf{u}^\nu - \mathbf{u}^\star\|_{L^1(\Omega)} \leq C_K\sqrt{\nu_{\mathrm{phys}}}, \tag{87}$$

where C_K depends on the initial TV bound M and physical time T_{phys}, but is independent of ε. *Explanation:* This bound follows directly from the classical theorem of Kuznetsov (1976) under A0, with $C_K = C\sqrt{T_{\mathrm{phys}}} \cdot \mathrm{TV}(\mathbf{u}_0) \leq C\sqrt{T_{\mathrm{phys}}} \cdot M$.

Proof. The proof proceeds in four stages:

Stage 1. Using A1 and A2, construct a function-space decomposition and analyze the well-posedness of the BV-aware parameterization.

Stage 2. Based on one-sided Lipschitz analysis and a modified Gronwall inequality, using A2, A3, A4 to establish the trajectory error bound.

Stage 3. Introduce the physical PDE framework and Kruzhkov theory, stating the BV compactness required for Stage 4.

Stage 4. Using A0, A5, A6 to assemble the Wasserstein error, obtaining $W_1 \leq C_{\mathrm{final}}\varepsilon^{1/2}$.

Notation (unified for the entire proof)

State space (diffusion model domain)

$u \in \mathbb{R}^d$ State vector ($d = N_x$ is the number of spatial grid points)

$\tau \in [0, T_d]$ Diffusion time (reverse: $T_d \to 0$, i.e., noise $\to$ data)

$p(u, \tau)$ Smoothed state density

$s(u, \tau) = \nabla_u \log p(u, \tau)$ True score field

$s_\theta(u, \tau)$ Network-approximated score field

$e(u, \tau) = s_\theta(u, \tau) - s(u, \tau)$ Score residual field

$\Sigma_{\text{score}}(\tau) \subset \mathbb{R}^d$ Shock manifold

$d_\Sigma(u, \tau)$ Signed distance function to $\Sigma_{\text{score}}(\tau)$

Physical space (PDE domain)

$x \in \Omega \subset \mathbb{R}$ Physical space coordinate

$t \in [0, T]$ Physical time (evolved by the PDE from initial data, orthogonal to τ)

$\mathbf{u}^\star(x)$ Kruzhkov entropy solution (at fixed $t = T_{\text{phys}}$)

$\mathbf{u}^\nu(x)$ Viscous PDE approximation with viscosity ν_{phys} (at $t = T_{\text{phys}}$)

∂_x Physical space partial derivative

$\|\cdot\|_{L^1(\Omega)}, \|\cdot\|_{L^2(\Omega)}$ L^p norms on physical space

Trajectories and generation

$\widehat{u}(\tau)$ ODE trajectory driven by the learned score ($\mathbb{R}^d$-valued)

$u^\natural(\tau)$ ODE trajectory driven by the true score ($\mathbb{R}^d$-valued)

$u^\theta \coloneqq \widehat{u}(0)$ Final generated sample (at $\tau = 0$)

$\mu_\theta \coloneqq \text{Law}(u^\theta)$ Distribution of the generated samples

$\delta_{\mathbf{u}^\star}$ Dirac measure at the target Kruzhkov entropy solution

Norms and distances

$\|v\|$ Euclidean norm on state space $\mathbb{R}^d$

$\langle \cdot, \cdot \rangle$ $\mathbb{R}^d$ inner product

$\text{TV}(f) = \int_\Omega |\partial_x f| \, dx$ Total variation seminorm

$W_1(\mu, \nu)$ 1-Wasserstein distance (Kantorovich–Rubinstein dual)

Time mapping The physical meaning of diffusion time τ decreasing from T_d to 0:

- $\tau = T_d$: pure noise (prior, e.g., standard Gaussian)
- $\tau = 0$: data sample (corresponding to the PDE solution at physical time $t = T_{\text{phys}}$)
- **Viscosity matching**: $\sigma(\tau)^2(\tau) = 2\nu_{\text{phys}}\tau$, so the evolution of τ is equivalent to a physical process with viscosity ν_{phys}; the drift coefficient of the reverse ODE is ν_{phys}
- In Stage 3, the target $\mathbf{u}^\star$ is the PDE entropy solution at a *fixed physical time* $t = T_{\text{phys}}$ (T_{phys} is regarded as a constant and is not written explicitly in the sequel)
- Diffusion time τ and physical time t are *orthogonal*: τ describes the noising–denoising intensity, t describes the PDE evolution

A.5 Stage 1: Function Space and Error Localization

A.5.1 1.1 Setup

Let the dimensionless diffusion time of the target distribution be $\tau \in (\tau_{\text{end}}, T_d]$. Denote the true smoothed data density by

$$p(u,\tau) = (\rho * G_\tau)(u), \qquad G_\tau(u) = (4\pi\tau)^{-d/2} \exp\bigl(-\|u\|^2/4\tau\bigr), \tag{88}$$

where $\rho(u)$ is the target data distribution. The true score field:

$$s(u,\tau) = \nabla_u \log p(u,\tau). \tag{89}$$

Define the shock manifold $\Sigma_{\text{score}}(\tau) \subset \mathbb{R}^d$ as the equipotential surface in state space that connects different modes. For any $u \in \mathbb{R}^d$, define its signed distance to $\Sigma_{\text{score}}(\tau)$:

$$d_\Sigma(u,\tau) = \inf_{v \in \Sigma_{\text{score}}(\tau)} \|u - v\| \cdot \operatorname{sgn}(\text{mode label}). \tag{90}$$

The shock interfacial layer neighborhood and the outer smooth region:

$$\Omega_{\text{sh}}(\tau) = \bigl\{u \in \mathbb{R}^d \;\bigm|\; |d_\Sigma(u,\tau)| \le \mathcal{O}(\tau)\bigr\}, \qquad \Omega_{\text{out}}(\tau) = \mathbb{R}^d \setminus \Omega_{\text{sh}}(\tau). \tag{91}$$

A.5.2 1.2 Lemma 1.1: Asymptotic Expansion and Singularity Decomposition of the True Score Field

Lemma 16. *Under Assumption A1, the true score field $s(u,\tau)$ can be decomposed on the whole space $\mathbb{R}^d$ as*

$$s(u,\tau) = s^{\text{sm}}(u,\tau) + s^{\text{sing}}(u,\tau) + \mathcal{O}(\tau), \tag{92}$$

where:

1. *$s^{\text{sm}}(u,\tau)$ is uniformly Lipschitz continuous on the whole space: $\|\nabla_u s^{\text{sm}}\|_{\text{op}} \le L_{\text{sm}} = \mathcal{O}(1)$.*
2. *$s^{\text{sing}}(u,\tau)$ is non-negligible only inside $\Omega_{\text{sh}}(\tau)$, and has the exact form*

$$s^{\text{sing}}(u,\tau) = \frac{\Delta s_{\text{jump}}(u)}{2} \tanh\Bigl(\frac{A(u,\tau) - B(u,\tau)}{2\tau}\Bigr), \tag{93}$$

where A, B are the potential functions of the dominant modes and $\Delta s_{\text{jump}} = \nabla_u B - \nabla_u A$ is the jump intensity vector.

Proof sketch (Laplace method): By A1, inside $\Omega_{\text{sh}}(\tau)$, the integral representation of $p(u,\tau)$ is dominated by the two nearest modes:

$$p(u,\tau) \sim C_A(u) \exp\Bigl(-\frac{A(u,\tau)}{\tau}\Bigr) + C_B(u) \exp\Bigl(-\frac{B(u,\tau)}{\tau}\Bigr). \tag{94}$$

Taking the log-gradient and using the identity $\nabla_u \log(e^{-A/\tau} + e^{-B/\tau})$ extracts the $\tanh$ part; the slowly varying coefficients C_A, C_B and their gradients are absorbed into s^{sm}. In $\Omega_{\text{out}}(\tau)$, one mode is overwhelmingly dominant, $\tanh(\cdot) \to \pm 1$, $\nabla_u s^{\text{sing}}$ tends to 0, and the score field degenerates to the smooth $\nabla_u A$ or $\nabla_u B$. The error term $\mathcal{O}(\tau)$ originates from the next-order remainder of the Laplace method; the uniform estimate is guaranteed by the mode-separation condition of A1. ■

Remark 17 (Nature of the singularity). Under standard parameterization, if the network directly fits $s(u,\tau)$, the Jacobian $\nabla_u s^{\text{sing}}$ of s^{sing} contains a singular component of $\mathcal{O}(1/\tau)$ inside Ω_{sh}—this is precisely the origin of the global Lipschitz constant $L(\tau) \propto 1/\tau$ in classical theory, which causes the Gronwall integral to diverge and produces the exponential amplification factor $\exp(\Lambda T_d)$. The core idea of parameterization (4) is to *explicitly embed* this singularity into the network structure, so that the residual field $e(u,\tau)$ no longer inherits the $1/\tau$ singularity.

A.5.3 1.3 Lemma 1.2: Well-Posedness of the BV-Aware Parameterization

The parameterization (4) defines the network score field:

$$s_\theta(u,\tau) = \underbrace{\nabla_u \phi_\theta^{\mathrm{sm}}(u,\tau)}_{\text{smooth background term}} + \underbrace{\frac{\kappa_\theta(u,\tau)}{2}\tanh\Big(\frac{D_\theta(u,\tau)}{2\tau}\Big)\nabla_u D_\theta(u,\tau)}_{\text{interfacial singular term}}. \tag{95}$$

Lemma 18. *(Spatial alignment and network well-posedness) Under Assumption A2, the map* $\Theta : (u,\tau) \mapsto (\phi_\theta^{\mathrm{sm}}, \kappa_\theta, D_\theta)$ *has the following properties:*

1. *Target approximation:* $\nabla_u \phi_\theta^{\mathrm{sm}} \approx s^{\mathrm{sm}}$, $\kappa_\theta \nabla_u D_\theta \approx \Delta s_{\mathrm{jump}}$, $D_\theta \approx A - B$.
2. *Bounded Jacobian: The Jacobian norms* $\|\nabla_u^2 \phi_\theta^{\mathrm{sm}}\|_{\mathrm{op}}$, $\|\nabla_u \kappa_\theta\|$, $\|\nabla_u^2 D_\theta\|_{\mathrm{op}}$ *are all independent of* τ*, of order* $\mathcal{O}(1)$.
3. *The network learns only the smooth signed distance field* D_θ*. The singular factor* $1/\tau$ *is written as an explicit physical prior* outside *the network structure, appearing only inside* $\tanh$ *and in the singular Jacobian matrix* J_{sh} *after differentiation; the network involves no* $1/\tau$ *amplification from automatic differentiation, completely avoiding gradient explosion.*

Proof: By A2, $\nabla_u^2 \phi_\theta^{\mathrm{sm}}$, $\nabla_u^2 D_\theta$, $\nabla_u \kappa_\theta$ are all uniformly bounded. Note that the derivative of $\tanh$ with respect to its argument is $\mathrm{sech}^2(\cdot) \le 1$, so when computing ∇_u of s_θ, the dependence of τ in the denominator does not propagate divergently outward; it passes into the Jacobian and forms a strictly negative semi-definite singular part (proved in detail below). Since A and B are both $\mathcal{O}(1)$ potential functions, the network-learned $D_\theta(u,\tau)$ has gradient $\nabla_u D_\theta = \mathcal{O}(1)$, introducing no additional singular scale. ■

A.5.4 1.4 Boundedness of the Residual Field (Heuristic Discussion)

Remark 19 (Heuristic significance of Corollary 1.4; not a rigorous claim). The following is a heuristic argument for intuition only and is *not relied upon in subsequent steps of the proof.*

Define the score residual field $e(u,\tau) = s_\theta(u,\tau) - s(u,\tau)$, decomposed into $e = e_{\mathrm{sm}} + e_{\mathrm{sh}}$, where e_{sh} is the shock residual arising from interface location/intensity deviation.

In the standard architecture, e_{standard} cannot fit the steep gradient of $\tanh(\cdot/\tau)$ and itself has a gradient component of $-\mathcal{O}(1/\tau)$ inside Ω_{sh}, causing exponential error amplification. The BV-aware architecture embeds the $\tanh$ structure, making the error of e_{sh} manifest as a small shift of the shock center position; this shift is of the same order as the shift magnitude in the W_1 sense ($\Delta W_1 \propto \Delta x$), without involving multiplicative amplification of gradients.

More rigorously, if one can independently prove gradient bounds for e_{sm} and for the shock shift of e_{sh}, one can establish $\int_0^{T_d} \|\nabla_u e(u,\tau)\|_{\mathrm{eff}}\, d\tau \le C_0 = \mathcal{O}(1)$. Rigorization of this direction is left to future work (see technical point C1.4-G in Appendix C). In the present proof, the Gronwall argument of Stage 2 *does not rely on this bound* and uses only the integral estimate of $\|e\|$ itself (see Theorem 2.2).

A.6 Stage 2: Modified Gronwall Inequality and Trajectory Error Bound

A.6.1 2.1 Failure of the Classical Analysis and the Gronwall Trap

In the reverse generative process, consider two ODE trajectories—the reference trajectory $u^\natural(\tau)$ driven by the true score s and the approximate trajectory $\widehat{u}(\tau)$ driven by the network score s_θ. Both obey the probability-flow ODE under the viscosity matching strategy $\sigma(\tau)^2(\tau) = 2\nu_{\mathrm{phys}}\tau$ (drift coefficient $\sigma\dot{\sigma} = \nu_{\mathrm{phys}}$):

$$\frac{du^\natural}{d\tau} = -\nu_{\mathrm{phys}}\, s(u^\natural,\tau) =: V(u^\natural,\tau), \qquad \frac{d\widehat{u}}{d\tau} = -\nu_{\mathrm{phys}}\, s_\theta(\widehat{u},\tau) =: V_\theta(\widehat{u},\tau). \tag{96}$$

Initial condition (A4). By A4, the error between the two trajectories at $\tau = T_d$ satisfies

$$E(T_d) := \|\widehat{u}(T_d) - u^\natural(T_d)\| \le \tilde{C}\varepsilon, \tag{97}$$

(equal to 0 if the prior matches exactly; small deviations are handled uniformly by the Gronwall formula below).

Define the trajectory error:

$$E(\tau) \coloneqq \|\widehat{u}(\tau) - u^\natural(\tau)\|. \tag{98}$$

Differentiating E^2:

$$\frac{1}{2}\frac{d}{d\tau}E(\tau)^2 = \big\langle \widehat{u} - u^\natural,\ V_\theta(\widehat{u},\tau) - V(u^\natural,\tau)\big\rangle. \tag{99}$$

Decompose the right-hand side into the self-consistency term (of V_θ) and the error-driven term:

$$\begin{aligned}\frac{1}{2}\frac{d}{d\tau}E(\tau)^2 \le\ & \underbrace{\big\langle \widehat{u} - u^\natural,\ V_\theta(\widehat{u},\tau) - V_\theta(u^\natural,\tau)\big\rangle}_{\text{Term I: drift self-consistency}} \\ & + \underbrace{\nu_{\text{phys}}\big\langle \widehat{u} - u^\natural,\ e(u^\natural,\tau)\big\rangle}_{\text{Term II: score error-driven}}.\end{aligned} \tag{100}$$

Applying Cauchy–Schwarz to Term II: $\nu_{\text{phys}}\langle \widehat{u} - u^\natural, e(u^\natural,\tau)\rangle \le \nu_{\text{phys}}\|e(u^\natural,\tau)\| \cdot E(\tau)$. Thus:

$$\frac{1}{2}\frac{d}{d\tau}E(\tau)^2 \le \underbrace{\big\langle \widehat{u} - u^\natural,\ V_\theta(\widehat{u},\tau) - V_\theta(u^\natural,\tau)\big\rangle}_{\text{Term I}} + \nu_{\text{phys}}\|e(u^\natural,\tau)\| \cdot E(\tau). \tag{101}$$

In the classical proof, Term I is bounded using the global Lipschitz constant $L_\theta(\tau) = \mathcal{O}(1/\tau)$, causing the Gronwall integral to diverge as $\exp(\Lambda T_d)$. Below we replace this with a one-sided Lipschitz constant.

A.6.2 2.2 Lemma 2.1: One-Sided Lipschitz Condition and Negative Semi-Definiteness

Definition 20 (One-sided Lipschitz constant)**.** The one-sided Lipschitz constant $\mu(\tau)$ of V_θ is the smallest number such that for all $u, v \in \mathbb{R}^d$:

$$\big\langle u - v,\ V_\theta(u,\tau) - V_\theta(v,\tau)\big\rangle \le \mu(\tau)\|u - v\|^2. \tag{102}$$

When V_θ is differentiable, $\mu(\tau)$ equals the maximum eigenvalue of the symmetric part of its Jacobian $J_\theta(u,\tau) \coloneqq \nabla_u V_\theta(u,\tau)$:

$$\mu(\tau) = \sup_{u\in\mathbb{R}^d} \lambda_{\max}\Big(\frac{J_\theta(u,\tau) + J_\theta(u,\tau)^\top}{2}\Big). \tag{103}$$

Lemma 21. *(Bounded logarithmic norm of the BV-aware architecture) Under Assumption A2,*

$$\mu(\tau) \le L_{\text{sm}} = \mathcal{O}(1), \tag{104}$$

i.e., the one-sided Lipschitz constant is independent of τ and has no singularity.

Proof: Compute the Jacobian of $V_\theta(u,\tau) = -\nu_{\text{phys}} s_\theta(u,\tau)$:

$$J_\theta = \nabla_u V_\theta = J_{\text{sm}} + J_{\text{sh}} + J_{\text{cross}}, \tag{105}$$

with the following components.

(a) Smooth part. $J_{\text{sm}} = -\nu_{\text{phys}}\nabla_u^2\phi_\theta^{\text{sm}}$. By A2, $\|\nabla_u^2\phi_\theta^{\text{sm}}\|_{\text{op}} = \mathcal{O}(1)$, so $\lambda_{\max}(J_{\text{sm}}) \le \nu_{\text{phys}} \cdot \mathcal{O}(1) = \mathcal{O}(1)$ (since $\nu_{\text{phys}} = \Theta(\varepsilon)$ is small but $\varepsilon \le 1$, so it does not exceed $\mathcal{O}(1)$).

(b) Cross terms. J_{cross} contains derivative terms of κ_θ and $\nabla_u D_\theta$. By A2 these are all uniformly bounded; moreover, the dominant terms produced by these derivatives are already controlled by explicit separation and internal derivative control, introducing no $1/\tau$ factor into this term, so $\lambda_{\max}(J_{\text{cross}}) = \mathcal{O}(1)$.

(c) Singular part. Differentiating the $\tanh$ term in the parameterization with respect to ∇_u yields

$$J_{\text{sh}}(u,\tau) = -\nu_{\text{phys}} \cdot \frac{\kappa_\theta(u,\tau)}{4\tau}\ \operatorname{sech}^2\Big(\frac{D_\theta(u,\tau)}{2\tau}\Big)\ \big(\nabla_u D_\theta\big)\big(\nabla_u D_\theta\big)^\top. \tag{106}$$

The matrix $(\nabla_u D_\theta)(\nabla_u D_\theta)^\mathsf{T}$ is a rank-1 positive semi-definite matrix (of the form $\mathbf{n}\mathbf{n}^\mathsf{T}$, where $\mathbf{n} = \nabla_u D_\theta$ is the shock interface normal vector). By A2, $\kappa_\theta \geq \kappa_0 > 0$; together with $\nu_{\text{phys}} > 0$, $\tau > 0$, $\text{sech}^2(\cdot) > 0$, the leading negative sign makes J_{sh} *strictly negative semi-definite*.

Hence the maximum eigenvalue of the symmetric part of J_{sh} satisfies $\lambda_{\max}(J_{\text{sh}}) \leq 0$, contributing nothing positive to the one-sided Lipschitz constant.

(d) Physical interpretation. In the direction normal to the shock surface (the $\mathbf{n}$ direction), J_{sh} provides a negative eigenvalue of magnitude $-\nu_{\text{phys}}\kappa_0\|\nabla_u D_\theta\|^2/(4\tau)$, corresponding to a *strong attractive force* of the reverse diffusion flow in the shock normal direction. Errors that deviate from the true shock manifold are compressed at this enormous rate rather than amplified—this is the mathematical essence of Burgers characteristic line convergence, and the core guarantee that this architecture avoids exponential divergence.

Combining (a)–(d),

$$\mu(\tau) \leq \lambda_{\max}\big(J_{\text{sm}} + J_{\text{cross}}\big) + \underbrace{\lambda_{\max}(J_{\text{sh}})}_{\leq 0} \leq L_{\text{sm}} = \mathcal{O}(1). \tag{107}$$

■

A.6.3 2.3 Theorem 2.2: Modified Flow Stability with Constant Upper Bound

Lemma 22. *(Trajectory error bound) Under Assumptions A2, A3, A4, let ε_{raw} be the raw (unnormalized) score matching mean-squared error bound, i.e., $\mathbb{E}[\|e\|^2] \leq \varepsilon_{\text{raw}}^2$. By the grid-normalized definition of A3, $\mathbb{E}[\Delta x\|e\|^2] \leq \varepsilon^2$, so $\varepsilon_{\text{raw}}^2 = \varepsilon^2/\Delta x$. Then*

$$\mathbb{E}_z\big[E(0)\big] \leq C_1\varepsilon_{\text{raw}} = C_1\frac{\varepsilon}{\sqrt{\Delta x}}, \tag{108}$$

where C_1 depends only on $T_d, \nu_{\text{phys}}, L_{\text{sm}}$. In the norm conversion of Stage 4, $\sqrt{\Delta x}$ will be canceled, yielding a grid-independent final bound.

Proof: **Step 1: Scalar differential inequality.** Applying Lemma 2.1 to Term I in the derivative of $\frac{1}{2}E^2$ gives

$$\frac{1}{2}\frac{d}{d\tau}E(\tau)^2 \leq \mu(\tau)\,E(\tau)^2 + \nu_{\text{phys}}\,\|e(u^\natural,\tau)\| \cdot E(\tau). \tag{109}$$

Dividing by $E(\tau)$ (when $E(\tau) > 0$; the inequality holds trivially when $E(\tau) = 0$),

$$\frac{d}{d\tau}E(\tau) \leq \mu(\tau)\,E(\tau) + \nu_{\text{phys}}\,\|e(u^\natural,\tau)\|. \tag{$\star$}$$

Step 2: Apply Gronwall's inequality. From ($\star$), integrating from $\tau = T_d$ to $\tau = 0$,

$$E(0) \leq E(T_d) \cdot \exp\Big(\int_0^{T_d}\mu(r)\,dr\Big) + \int_0^{T_d}\exp\Big(\int_0^{\tau}\mu(r)\,dr\Big)\,\nu_{\text{phys}}\,\|e(u^\natural,\tau)\|\,d\tau. \tag{110}$$

By Lemma 2.1, $\mu(r) \leq L_{\text{sm}} = \mathcal{O}(1)$, so

$$\exp\Big(\int_0^{\tau}\mu(r)\,dr\Big) \leq e^{L_{\text{sm}}T_d} =: C_{\text{flow}} = \mathcal{O}(1). \tag{111}$$

Substituting the initial error bound $E(T_d) \leq \tilde{C}\varepsilon$ from A4:

$$E(0) \leq \tilde{C}C_{\text{flow}}\,\varepsilon + C_{\text{flow}}\,\nu_{\text{phys}}\int_0^{T_d}\|e(u^\natural,\tau)\|\,d\tau. \tag{$\star\star$}$$

Step 3: Integral bound of the score matching error (directly from A3). Apply Cauchy–Schwarz to the integral $\int_0^{T_d}\|e(u^\natural,\tau)\|\,d\tau$:

$$\int_0^{T_d}\|e(u^\natural,\tau)\|\,d\tau \leq \sqrt{T_d}\cdot\Big(\int_0^{T_d}\|e(u^\natural,\tau)\|^2\,d\tau\Big)^{1/2}. \tag{112}$$

Take expectation over z (prior noise) of the right-hand side. By A3, $\mathbb{E}[\Delta x\|e\|^2] \le \varepsilon^2$, i.e., the raw (unnormalized) mean-squared error satisfies $\mathbb{E}[\|e\|^2] \le \varepsilon^2/\Delta x = \varepsilon_{\text{raw}}^2$. Hence

$$\mathbb{E}_z\Big[\int_0^{T_d} \|e(u^\natural,\tau)\|^2\, d\tau\Big] \le T_d \cdot \varepsilon_{\text{raw}}^2 = T_d \cdot \frac{\varepsilon^2}{\Delta x}, \tag{113}$$

(this step holds because the marginal distribution of $u^\natural(\tau)$ is $p(\cdot,\tau)$, so the expectation over $(u^\natural(\tau),\tau)$ matches $\mathbb{E}_{u,\tau}$ in A3).

Now take expectation of the Cauchy–Schwarz result (using Jensen's inequality):

$$\mathbb{E}_z\Big[\int_0^{T_d} \|e(u^\natural,\tau)\|\, d\tau\Big] \le \sqrt{T_d} \cdot \sqrt{T_d\, \varepsilon_{\text{raw}}^2} = T_d\, \varepsilon_{\text{raw}} = T_d \frac{\varepsilon}{\sqrt{\Delta x}}. \tag{114}$$

Step 4: Assembly. Substituting Step 3 into $(\star\star)$ and taking expectation:

$$\mathbb{E}_z[E(0)] \le \tilde{C} C_{\text{flow}}\, \varepsilon_{\text{raw}} + C_{\text{flow}}\, \nu_{\text{phys}}\, T_d\, \varepsilon_{\text{raw}} =: C_1 \varepsilon_{\text{raw}}, \tag{115}$$

where $C_1 = C_{\text{flow}}(\tilde{C} + \nu_{\text{phys}} T_d) = \mathcal{O}(1)$ (by A5, $\nu_{\text{phys}} = \Theta(\varepsilon) \le 1$, so $\nu_{\text{phys}} T_d = \mathcal{O}(1)$). From $\varepsilon_{\text{raw}} = \varepsilon/\sqrt{\Delta x}$, we obtain $\mathbb{E}_z[E(0)] \le C_1 \varepsilon/\sqrt{\Delta x}$. ■

Conclusion. Under the BV-aware architecture, the trajectory endpoint error is $\mathbb{E}[E(0)] = \mathcal{O}(\varepsilon_{\text{raw}})$, free of the exponential amplification factor.

A.7 Stage 3: Kruzhkov Entropy Solution L^1 Contraction Framework

Remark 23 (Convention for Stage 3). We now introduce physical space $x \in \Omega \subset \mathbb{R}$. The components of the state vector $u \in \mathbb{R}^d$ ($d = N_x$) are $u_i = \mathbf{u}(x_i)$, where $\{x_i\}$ is a uniform grid in physical space (spacing $\Delta x = |\Omega|/N_x$). The discrete L^1 norm is defined as $\|\mathbf{u}\|_{L^1} = \Delta x \sum_{i=1}^{N_x} |\mathbf{u}(x_i)|$, which approximates the continuous L^1 norm on physical space (see the norm equivalence claim in Stage 4).

A.7.1 3.1 Kruzhkov Entropy Condition and L^1 Contraction

Consider the target hyperbolic conservation law (on physical space $\Omega \times [0,T]$):

$$\partial_t \mathbf{u} + \partial_x f(\mathbf{u}) = 0, \qquad x \in \Omega \subset \mathbb{R}, \quad t \in [0,T], \tag{116}$$

where f is a flux function (e.g., Burgers flux $f(u) = u^2/2$), with initial condition $\mathbf{u}(\cdot,0) = \mathbf{u}_0$. The Kruzhkov entropy solution $\mathbf{u}^\star$ satisfies: for every constant $k \in \mathbb{R}$, in the sense of distributions,

$$\partial_t |\mathbf{u}^\star - k| + \partial_x\big(\operatorname{sgn}(\mathbf{u}^\star - k)(f(\mathbf{u}^\star) - f(k))\big) \le 0. \tag{117}$$

Corollary of Kruzhkov's theorem (L^1-contractive semigroup). If $\mathbf{a}(x,t)$ and $\mathbf{b}(x,t)$ both satisfy the Kruzhkov entropy condition, then

$$\|\mathbf{a}(\cdot,t) - \mathbf{b}(\cdot,t)\|_{L^1(\Omega)} \le \|\mathbf{a}(\cdot,0) - \mathbf{b}(\cdot,0)\|_{L^1(\Omega)}. \tag{118}$$

Set $\mathbf{u}^\star(x) := \mathbf{u}^\star(x, T_{\text{phys}})$ (at fixed physical time T_{phys}; omitted in the sequel).

A.7.2 3.2 Loss Design and BV Compactness

The loss function $\mathcal{L}_3$ augments the standard denoising score matching loss with a BV regularizer:

$$\mathcal{L}_3 = \mathcal{L}_{\text{DSM}} + \lambda_{\text{BV}}\, \text{TV}(D_\theta). \tag{119}$$

By Assumption A0, together with the hard-coded $\tanh$ structure of parameterization (4), this regularizer guarantees that the generated samples almost surely belong to the bounded BV ball $\mathcal{K}_M$. By Helly's selection theorem, $\mathcal{K}_M$ is relatively compact in the $L^1(\Omega)$ topology.

A.7.3 3.3 Viscous Limit (Application of Kuznetsov's Lemma)

The reverse stage of the diffusion model, under viscosity matching $\sigma(\tau)^2(\tau) = 2\nu_{\text{phys}}\tau$, effectively introduces artificial viscosity ν_{phys}. Denote by $\mathbf{u}^\nu$ the solution of the viscous physical PDE at $t = T_{\text{phys}}$:

$$\partial_t \mathbf{u}^\nu + \partial_x f(\mathbf{u}^\nu) = \nu_{\text{phys}}\, \partial_{xx}\mathbf{u}^\nu. \tag{120}$$

By Assumption A6 (based on Kuznetsov 1976):

$$\|\mathbf{u}^\nu - \mathbf{u}^\star\|_{L^1(\Omega)} \le C_K \sqrt{\nu_{\text{phys}}}. \tag{121}$$

Together with A5 ($\nu_{\text{phys}} = \Theta(\varepsilon)$), we obtain

$$\|\mathbf{u}^\nu - \mathbf{u}^\star\|_{L^1(\Omega)} \le C_K\sqrt{C\varepsilon} = C_K\sqrt{C}\cdot\varepsilon^{1/2}. \tag{3.1}$$

A.8 Stage 4: Assembly of the Wasserstein Error and Emergence of $\mathcal{O}(\varepsilon^{1/2})$

A.8.1 4.1 Grid Renormalization, Norm Equivalence, and Neural Generation Error

Derivation of the discrete L^1–normalized Euclidean norm equivalence. The physical domain Ω is discretized into $d = N_x$ uniform grid points with spacing $\Delta x = |\Omega|/d$. For a state vector $u \in \mathbb{R}^d$, the discrete L^1 norm is

$$\|\mathbf{u}\|_{L^1} = \Delta x \sum_{i=1}^{d} |u_i|. \tag{122}$$

From the ℓ^1–ℓ^2 norm inequality ($\sum |u_i| \le \sqrt{d}\sqrt{\sum u_i^2} = \sqrt{d}\|u\|$), substituting Δx, we obtain

$$\|\mathbf{u}\|_{L^1} \le \Delta x\sqrt{d}\,\|u\| = \sqrt{|\Omega|}\cdot\sqrt{\Delta x}\,\|u\|. \tag{4.1}$$

Deriving the L^1 bound for the neural generation error. Note that in the renormalized Assumption A3, the error bound is given in the form $\mathbb{E}[\Delta x\|e\|^2] \le \varepsilon^2$, which means that for the pure Euclidean norm, we have $\mathbb{E}[\|e\|^2] \le \varepsilon^2/\Delta x$.

Recall from Stage 2, the Gronwall integration and trajectory error bound Lemma 2.2 give, for the standard Euclidean distance $E(0)$:

$$\mathbb{E}_z\big[\|\widehat{u}(0) - u^\natural(0)\|\big] \le C_1\,\mathbb{E}_z\big[\|e\|\big] \le C_1\frac{\varepsilon}{\sqrt{\Delta x}}. \tag{4.2}$$

The corresponding difference of physical functions is $u^\theta - \mathbf{u}^\nu$. Combining with the norm scaling relation (4.1):

$$\mathbb{E}_z\big[\|u^\theta - \mathbf{u}^\nu\|_{L^1(\Omega)}\big] \le \sqrt{|\Omega|}\sqrt{\Delta x}\cdot\mathbb{E}_z\big[\|\widehat{u}(0) - u^\natural(0)\|\big] \le \sqrt{|\Omega|}\sqrt{\Delta x}\cdot C_1\frac{\varepsilon}{\sqrt{\Delta x}}. \tag{123}$$

Here d and Δx are *perfectly canceled*! Regardless of how fine the grid is, the error bound remains valid, yielding a grid-independent continuum limit bound:

$$\mathbb{E}_z\big[\|u^\theta - \mathbf{u}^\nu\|_{L^1(\Omega)}\big] \le C_1\sqrt{|\Omega|}\,\varepsilon =: C_2\varepsilon. \tag{4.3}$$

Remark 24 (Place of the Gagliardo–Nirenberg interpolation). An earlier draft used a Gagliardo–Nirenberg-type interpolation inequality $\|h\|_{L^1} \le C_{\text{GN}}\|h\|_{L^2}^{2/3}\text{TV}(h)^{1/3}$ to convert L^2 error into L^1 error, obtaining an $\mathcal{O}(\varepsilon^{2/3})$ bound. The present revised proof uses normalized L^2-matching combined with the norm equivalence (4.1), obtaining a sharper, dimension-independent L^1 error $\mathcal{O}(\varepsilon)$ directly from the scaled Euclidean error.

A.8.2 4.2 Triangle Inequality Decomposition of the Wasserstein Distance

The 1-Wasserstein distance (Kantorovich–Rubinstein dual):

$$W_1(\mu,\nu) = \inf_{\pi\in\Pi(\mu,\nu)} \int \|\mathbf{u} - \mathbf{v}\|_{L^1(\Omega)}\, d\pi(\mathbf{u},\mathbf{v}). \tag{124}$$

Let $\delta_{\mathbf{u}^\nu}$ be the Dirac measure at the viscous solution (given the initial data $\mathbf{u}_0$, the viscous PDE solution is unique, hence deterministic). Decompose via the triangle inequality:

$$W_1(\mu_\theta, \delta_{\mathbf{u}^\star}) \leq \underbrace{W_1(\mu_\theta, \delta_{\mathbf{u}^\nu})}_{\text{Neural generation error (Term A)}} + \underbrace{W_1(\delta_{\mathbf{u}^\nu}, \delta_{\mathbf{u}^\star})}_{\text{Viscous limit error (Term B)}}. \tag{4.4}$$

Term A (Neural generation error). Since $\widehat{u}(0)$ and $u^\natural(0)$ are deterministically generated from the same prior noise z (the ODE has a unique solution for fixed z), there exists a natural pointwise coupling:

$$\pi_z = \big(u^\theta(z),\ \mathbf{u}^\nu\big)_{\#}\, p_{T_d}, \tag{125}$$

where p_{T_d} is the prior distribution and $\mathbf{u}^\nu$ is regarded as a constant vector (independent of z). By the feasibility of this coupling and (4.3):

$$W_1(\mu_\theta, \delta_{\mathbf{u}^\nu}) \leq \int \|u^\theta(z) - \mathbf{u}^\nu\|_{L^1(\Omega)}\, dp_{T_d}(z) = \mathbb{E}_z\big[\|u^\theta - \mathbf{u}^\nu\|_{L^1(\Omega)}\big] \leq C_2\varepsilon. \tag{4.5}$$

Term B (Viscous limit error). Both $\delta_{\mathbf{u}^\nu}$ and $\delta_{\mathbf{u}^\star}$ are single-point distributions; W_1 degenerates to the L^1 distance:

$$W_1(\delta_{\mathbf{u}^\nu}, \delta_{\mathbf{u}^\star}) = \|\mathbf{u}^\nu - \mathbf{u}^\star\|_{L^1(\Omega)}. \tag{126}$$

From (3.1) (the joint conclusion of A6 and A5):

$$W_1(\delta_{\mathbf{u}^\nu}, \delta_{\mathbf{u}^\star}) \leq C_K\sqrt{C} \cdot \varepsilon^{1/2}. \tag{4.6}$$

A.8.3 4.3 Final Conclusion

Substituting (4.5) and (4.6) into (4.4):

$$W_1(\mu_\theta, \delta_{\mathbf{u}^\star}) \leq C_2\varepsilon + C_K\sqrt{C} \cdot \varepsilon^{1/2}. \tag{127}$$

For $\varepsilon \in (0, 1)$, we have $\varepsilon \leq \varepsilon^{1/2}$, so the higher-order term $\mathcal{O}(\varepsilon)$ is absorbed by the lower-order term $\mathcal{O}(\varepsilon^{1/2})$:

$$W_1(\mu_\theta, \delta_{\mathbf{u}^\star}) \leq \big(C_2 + C_K\sqrt{C}\big) \cdot \varepsilon^{1/2} =: C_{\text{final}} \cdot \varepsilon^{1/2}. \tag{128}$$

Here $C_{\text{final}} = C_2 + C_K\sqrt{C}$ depends only on $M, T_d, T_{\text{phys}}, \kappa_0, L_{\text{sm}}, \nu_{\text{phys}}/\varepsilon$, all independent of ε. Thus

$$\boxed{W_1(\mu_\theta, \delta_{\mathbf{u}^\star}) \leq C_{\text{final}}\, \varepsilon^{1/2}.} \tag{129}$$

This completes the proof of Theorem 3. □

Appendix A: Symbol Correspondence Table (Old → Revised)

Table 3: Symbol correspondence table

Old symbol	Revised symbol
$u \in \mathbb{R}^d$ (state point) and $u(x,t)$ (PDE solution) confused	State space: $u \in \mathbb{R}^d$; physical space: $\mathbf{u}(x)$
u^θ used for both true trajectory and PDE solution	$u^\natural(\tau)$ (true trajectory); $\mathbf{u}^\star$ (PDE solution)
$\widehat{u}$ confused as trajectory/sample/distribution	$\widehat{u}(\tau)$ (trajectory); u^θ (sample); μ_θ (distribution)
∇_u vs ∂_x confused	∇_u (state-space gradient); ∂_x (physical space partial derivative)
$\sup_u \lambda_{\max}$ with ambiguous scope	$\sup_{u\in\mathbb{R}^d} \lambda_{\max}$
$\|E(\tau)\|$ with ambiguous norm type	$E(\tau) := \|\widehat{u}(\tau) - u^\natural(\tau)\|$ ($\mathbb{R}^d$ Euclidean norm)
Dual role of u in residual field $e(u,\tau)$	The parameter u in $e(u,\tau)$ is a state-space point
Relation between times τ and t unclear	Time mapping paragraph explicitly defined

Table 4: Summary of modifications

No.	Modification	Treatment
M1	Rigorous gradient bound claim in Corollary 1.4	Downgraded to heuristic remark in §A.5; not relied upon s
M2	The $\int \|e\| \, d\tau$ bound in Lemma 2.2	Directly from A3 + Cauchy–Schwarz; no dependence on C
M3	GN interpolation in Stage 4	Removed; replaced with discrete norm equivalence
M4	Viscous error Term B	Directly cites A5 + A6 (Equation (3.1)); logical chain
M5	Theorem statement	Explicitly references A0–A6; no dangling assumpt
M6	Assumption system	Newly added A0–A6, with verifiability annotations and engineering

Table 5: Checklist of technical points

No.	Technical point	Status	Scope of influence
C1.1-P	"Finitely separated Gaussian clusters" assumption of Lemma 1.1	Upgraded to A1	If A1 fails, the score singula
C1.4-G	Residual gradient bound of Corollary 1.4	Left to future work	This proof does not rely on
C2.1-κ	Training-time enforcement of $\kappa_\theta > 0$	Upgraded to A2	If not strictly enforced, conc
EA1	Applicability of A1 to general continuous distributions	Engineering approximation	Affects the uniformity of th
EA2	Feasibility of schedule $\nu_{\text{phys}} = \Theta(\varepsilon)$ in A5	Engineering approximation	If $\nu_{\text{phys}} \gg \varepsilon$, Term B domi
P4.1-GN	GN interpolation argument in the continuum limit	Retained in remark	Does not affect conclusion
K3.3-ν	Generalization of Kuznetsov bound in A6	Rigorously established	Only requires initial BV bo

Appendix B: Summary of Modifications

Appendix C: Technical Points Requiring Further Rigorization and Scope of Engineering Approximations

A.9 Proof of Theorem 5 (Shock-location admissibility)

Theorem 25 (Shock-location admissibility)**.** *In the limit $\sigma(\tau) \to 0$ (i.e., the final step of inference), the solution produced by the proposed sampler satisfies the Rankine–Hugoniot condition at each shock location $x_s(t)$ almost everywhere,*

$$\dot{x}_s(t) = \frac{f(u_L) - f(u_R)}{u_L - u_R}, \tag{130}$$

and the Lax entropy condition $f'(u_L) \geq \dot{x}_s \geq f'(u_R)$.

Proof. **Assumptions.**

1. **(Perfect score matching at the interfacial layer.)** The BV-aware score parameterization (4) is optimized to global optimality in a neighborhood of the discontinuity, i.e. $s_\theta(u, \tau) \equiv \nabla_u \log p_\tau(u)$. We thus analyze the true score field.

2. **(Single-shock target manifold.)** The target distribution $\rho_t(u)$ concentrates on a submanifold $\mathcal{M}_t$ in function space (or its discretization $\mathbb{R}^N$).For the sampling trajectory under consideration, the corresponding target Kruzhkov entropy solution $\mathbf{u}^\star(x, t)$ at the current time t possesses a single isolated shock located at $x_s(t)$, with left and right limits

$$u_L \coloneqq \mathbf{u}^\star(x_s^-, t), \qquad u_R \coloneqq \mathbf{u}^\star(x_s^+, t). \tag{131}$$

3. **(Local planarity in high dimension.)** Since the grid dimension N is large, in the limit $\tau \to 0$ the separating manifold of data modes (i.e., the locus where the score exhibits a shock) can be locally approximated as a hyperplane. Its signed distance function $\phi_\theta^{\text{sh}}(u, \tau)$ satisfies $|\nabla_u \phi_\theta^{\text{sh}}| = 1$.

4. **(Compressive score gradient.)** When the reverse sampling trajectory $u(\tau)$ approaches the target manifold $\mathcal{M}_t$, the local flow induced by the network parameterization s_θ is compressive along the shock normal $\nabla \phi_\theta^{\text{sh}}$:

$$\partial_{\phi_\theta^{\text{sh}}} s_\theta \cdot \nabla \phi_\theta^{\text{sh}} < 0. \tag{132}$$

This guarantees that the score field acts as a centripetal force, pushing probability mass toward the interfacial manifold.

Part I: Geometric correspondence via the viscosity-matched schedule.

Let the forward noising process be a variance-exploding SDE. Under the schedule (2), the noise variance is tied to the physical viscosity ν_{phys}:

$$\sigma(\tau)^2 = 2\nu_{\text{phys}}\,\tau, \qquad \tau \in [0, T_d]. \tag{133}$$

The forward SDE then reads

$$du = \sqrt{2\nu_{\text{phys}}}\,dw. \tag{134}$$

The corresponding Fokker–Planck equation for the density $p(u,\tau)$ in data space $\mathbb{R}^N$ is

$$\partial_\tau p(u,\tau) = \nu_{\text{phys}}\,\Delta_u p(u,\tau). \tag{135}$$

Differentiating the score $s(u,\tau) = \nabla_u \log p = \nabla_u p/p$ with respect to τ and invoking (135) gives the exact evolution of the score:

$$\begin{aligned}\partial_\tau s &= \nu_{\text{phys}}\,\nabla_u\Big(\frac{\Delta_u p}{p}\Big)\\ &= \nu_{\text{phys}}\,\Delta_u s + \nu_{\text{phys}}\,\nabla_u(|s|^2).\end{aligned} \tag{136}$$

Since s is a gradient field, $\nabla_u(|s|^2) = 2(s\cdot\nabla_u)s$. Note that the relation $s = \nabla_u \log p$ is essentially the classical *inverse Cole–Hopf transform*, which explicitly converts the linear heat equation (Fokker–Planck) into a nonlinear PDE. We thus obtain the **high-dimensional viscous Burgers equation at the score level**:

$$\partial_\tau s + (-2\nu_{\text{phys}}\,s\cdot\nabla_u)\,s = \nu_{\text{phys}}\,\Delta_u s. \tag{137}$$

Lemma 26 (Effective viscosity equivalence)**.** *Via the schedule* (2)*, the effective viscosity of the score dynamics is numerically identical to the physical viscosity* ν_{phys} *of the target equation.*

The target PDE with small physical viscosity is

$$\partial_t u(x,t) + \partial_x f(u(x,t)) = \nu_{\text{phys}}\,\partial_{xx} u(x,t). \tag{138}$$

Under discretization, the components of $u \in \mathbb{R}^N$ satisfy $u_i \approx u(x_i,t)$. The target distribution $\rho_t(u)$ (the boundary condition at $\tau = 0$) is the physical weak-solution distribution. In the limit $\tau \to 0$, the reverse sampling trajectory $u(\tau)$ is pulled by the score $s(u,\tau)$ toward the support manifold $\mathcal{M}_t$ of $\rho_t(u)$.

The BV-aware score parameterization (4) reads

$$s_\theta(u,\tau) = \nabla\phi_\theta^{\text{sm}} + \frac{\kappa_\theta}{2}\tanh\Big(\frac{\phi_\theta^{\text{sh}}}{2}\Big)\nabla\phi_\theta^{\text{sh}}. \tag{139}$$

Near the interface, the smooth background $\nabla\phi_\theta^{\text{sm}}$ is negligible, and the true score exhibits the asymptotic inner-scale form

$$s(u,\tau) \sim \frac{\kappa_\theta(u)}{2}\tanh\Big(\frac{\operatorname{dist}(u,\mathcal{M}_t)}{\sqrt{2\nu_{\text{phys}}\,\tau}}\Big)\mathbf{n}(u), \tag{140}$$

where $\mathbf{n}(u) = \nabla_u \operatorname{dist}(u,\mathcal{M}_t)$ is the normal vector to the manifold.

Lemma 27 (Geometric correspondence in the double limit)**.** *When* $\tau \to 0$ *and* $\nu_{\text{phys}} \to 0$ *hold simultaneously, the* $\tanh$ *profile in* (140) *degenerates to a step function. Consequently, the location where the score field exhibits a shock is determined in data space by* $\operatorname{dist}(u,\mathcal{M}_t) = 0$*. Mapped back to physical space, this manifold boundary coincides exactly with the grid coordinates where* $u(x,t)$ *exhibits a jump, i.e., the physical shock position* $x_s(t)$*.*

Part II: Weak-form limit and derivation of the Rankine–Hugoniot condition.

2.1 Tweedie profile near the shock.

For the reverse diffusion, the clean physical state $\widehat{u}_0(x,t)$ is estimated via the Tweedie formula

$$\widehat{u}_0(x,t) = u(\tau) + \sigma(\tau)^2\, s(u,\tau). \tag{141}$$

Under the schedule (2) and the BV-aware parameterization, restricted to a one-dimensional physical section, the signed distance expands as $\phi_\theta^{\mathrm{sh}} \approx \dfrac{x - x_s(t)}{\sqrt{2\nu_{\mathrm{phys}}\,\tau}}$.

Lemma 28 (Viscous shock profile)**.** *In a neighborhood of $x_s(t)$, the Tweedie-reconstructed physical field $\widehat{u}^\tau(x,t)$ possesses a smooth, monotone shock transition layer. To make the physical scales explicit, define the layer thickness $\varepsilon = \sigma(\tau) = \sqrt{2\nu_{\mathrm{phys}}\,\tau}$. It is described by*

$$\widehat{u}^\tau(x,t) \approx \frac{u_L + u_R}{2} - \frac{u_L - u_R}{2}\tanh\Big(\frac{x - x_s(t)}{2\varepsilon}\Big). \tag{142}$$

Thus $\widehat{u}^\tau$ is strictly between u_L and u_R and has controlled total variation.

2.2 Godunov guidance and weak form.

During generation, an entropy-compatible Godunov flux difference is used as PDE guidance. Thus, as $\tau \to 0$, the generated sequence $\widehat{u}^\tau$ approximately satisfies a weak form: for every smooth test function $\varphi \in C_c^\infty(\mathbb{R} \times (0,T))$,

$$I(\varepsilon) = \iint_{\mathbb{R}\times\mathbb{R}^+} \big(\widehat{u}^\tau\, \partial_t\varphi + f(\widehat{u}^\tau)\, \partial_x\varphi\big)\, dx\, dt = \mathcal{O}(\varepsilon). \tag{143}$$

Because the $\tanh$ profile guarantees $\widehat{u}^\tau \in L^\infty \cap \mathrm{BV}$, the dominated convergence theorem allows the limit $\varepsilon \to 0$ to be taken inside the integral. **This step is particularly emphasized: the local monotonicity induced by parameterization** (4) **is the key topological constraint that renders this weak limit legitimate.** Traditional diffusion models (e.g., those forcing a Gaussian fit across shock interfaces) inevitably produce Gibbs oscillations, causing the BV norm to blow up in the limit and precluding application of dominated convergence. The profile (142) converges strongly to the piecewise-constant function

$$\mathbf{u}^\star(x,t) = \begin{cases} u_L, & x < x_s(t), \\ u_R, & x > x_s(t). \end{cases} \tag{144}$$

Partition the spacetime domain along the shock trajectory $\Gamma : x = x_s(t)$ into a left region D^- ($x < x_s$) and a right region D^+ ($x > x_s$). Then

$$0 = \lim_{\varepsilon\to 0} I(\varepsilon) = \iint_{D^-} \big(u_L\, \partial_t\varphi + f(u_L)\, \partial_x\varphi\big)\, dx\, dt + \iint_{D^+} \big(u_R\, \partial_t\varphi + f(u_R)\, \partial_x\varphi\big)\, dx\, dt. \tag{145}$$

Inside each $D^\pm$, the solution u_L, u_R is a strong solution of the conservation law. Applying Green's theorem to transfer derivatives from φ onto u,

$$\begin{aligned} \iint_D \big(u\, \partial_t\varphi + f(u)\, \partial_x\varphi\big)\, dx\, dt = &\int_{\partial D} \varphi\, \big(u\, n_t + f(u)\, n_x\big)\, ds \\ &- \iint_D \varphi\, \big(\partial_t u + \partial_x f(u)\big)\, dx\, dt. \end{aligned} \tag{146}$$

Since the PDE holds pointwise in $D^\pm$, the area term vanishes. Because φ has compact support, only the boundaries along Γ contribute. On Γ, the outward normal (pointing toward D^+) is

$$\mathbf{n} = (n_x, n_t) = \frac{1}{\sqrt{1 + \dot{x}_s^2}}\big(1,\ -\dot{x}_s\big). \tag{147}$$

Accounting for orientation, the jump condition emerges:

$$\int_\Gamma \varphi \cdot \Big[\big(u_L(-\dot{x}_s) + f(u_L)\big) - \big(u_R(-\dot{x}_s) + f(u_R)\big)\Big] \frac{1}{\sqrt{1 + \dot{x}_s^2}}\, ds = 0. \tag{148}$$

Since φ is arbitrary along Γ, the algebraic bracket must vanish identically, yielding

$$\dot{x}_s\,(u_L - u_R) = f(u_L) - f(u_R). \tag{149}$$

For a genuine shock $u_L \neq u_R$, division by the difference gives the Rankine–Hugoniot condition

$$\dot{x}_s(t) = \frac{f(u_L) - f(u_R)}{u_L - u_R}. \qquad \square \tag{150}$$

Part III: Entropy compatibility and the Lax condition.

Background: the non-uniqueness of weak solutions. The Rankine–Hugoniot condition derived above is a *necessary* but not sufficient condition for a weak solution. Physically inadmissible "expansion shocks", which dissipate energy in an entropy-violating manner, also satisfy the R–H condition in nonlinear evolution models. Only by imposing a rigorous entropy condition—the Lax entropy condition, motivated by physical mechanisms such as the Double-Burgers coupling—do we ensure that the generated solution is not merely a "numerical fit" to data, but a genuine approximation of the *physically correct* conservation law.

3.1 Divergence–characteristic relation.

For a convex flux f (e.g., the Burgers flux $f(u) = u^2/2$), a physical shock must satisfy the Lax entropy condition

$$f'(u_L) > \dot{x}_s > f'(u_R). \tag{151}$$

In the reverse-time generative ODE,

$$\frac{du}{d\tau} = -\frac{1}{2}\,\beta(\tau)\big[u + s_\theta(u,\tau)\big], \tag{152}$$

Tweedie reconstruction yields $\partial_x \widehat{u} \approx \partial_x(u + \sigma(\tau)^2\, s_\theta) \to -\infty$ near x_s. This singular negative gradient corresponds to the compression of characteristics toward the shock in physical space.

3.2 Kruzhkov entropy inequality.

To prove that $\widehat{u}$ is an entropy solution, we verify the Kruzhkov inequality for every constant k (in the sense of distributions):

$$\partial_t\,|\widehat{u} - k| + \partial_x\big[\mathrm{sgn}(\widehat{u} - k)\,(f(\widehat{u}) - f(k))\big] \leq 0. \tag{153}$$

By Theorem 1 (Double-Burgers coupling), the score field itself satisfies a viscous Burgers equation. Hence the generative dynamics driven by s_θ constitute a JKO gradient flow that minimizes an energy functional. A jump $u_L \to u_R$ violating the Lax condition (i.e., $u_L < u_R$) would correspond to an expansion (rarefaction) fan whose characteristics diverge outward. Under the BV-aware parameterization (4), the sign of the score gradient is determined by the $\tanh$ profile.

3.3 Proof by contradiction.

Suppose a non-physical shock (an expansion shock) satisfies the R–H condition but has $u_L < u_R$.

- **(i) Score behavior.** The Tweedie formula then requires the score s_θ to supply an outward "push" in order to sustain this unstable discontinuity.
- **(ii) Burgers consistency loss.** The loss term $\mathcal{L}_{\mathrm{Burg}}$ (Burgers consistency) enforces $\partial_\tau s + (s \cdot \nabla)s = \nu_{\mathrm{phys}}\,\Delta s$. An expansion jump would be immediately regularized by the diffusion term Δs, producing a large residual in $\mathcal{L}_{\mathrm{Burg}}$.
- **(iii) Conclusion.** Minimizing the total loss systematically eliminates all non-physical jumps, because only compressive configurations satisfying the Lax condition (151) are compatible with the stable evolution operator of $\mathcal{L}_{\mathrm{Burg}}$.

3.4 Stability in the $\tau \to 0$ limit.

Finally, the Oleinik entropy condition provides the estimate near the shock:

$$\frac{\widehat{u}(x+a,t) - \widehat{u}(x,t)}{a} \leq \frac{C}{t}, \tag{154}$$

which is guaranteed by the controlled width of the $\tanh$ layer in parameterization (4). As τ decreases, the profile steepens but remains strictly monotone whenever $u_L > u_R$, thereby precluding non-physical oscillations.

Combining the three parts above, the generated solution satisfies both the Rankine–Hugoniot condition and the Lax entropy condition almost everywhere. □

A.10 Proof of Theorem 6 (JKO correspondence)

Theorem 29 (JKO correspondence)**.** *Let p_τ denote the density of the reverse sampler at diffusion time $\tau \in [0, T]$, and let $\rho^\star$ be the target distribution induced by the PDE solution in data space. Define the free energy functional*

$$\mathcal{F}(\rho) = \mathrm{Ent}(\rho \mid \rho^\star) + \lambda\, \mathcal{R}(\rho), \qquad \mathrm{Ent}(\rho \mid \rho^\star) = \int_{\mathbb{R}^N} \rho(x)\, \log \frac{\rho(x)}{\rho^\star(x)}\, dx, \tag{155}$$

where $\mathcal{R}$ is a density functional encoding the PDE constraint and $\lambda \geq 0$.

Assume the following hypotheses:

1. ***(Exact score limit.)*** *After sufficient training, $s_\theta(x, \tau) = \nabla_x \log p_\tau(x)$ pointwise.*
2. ***(Viscosity-matched schedule.)*** *$a(\tau) = g(\tau)^2/2 > 0$, aligned with the physical viscosity.*
3. ***(PDE constraint as a differentiable functional.)*** *$\mathcal{R}$ is Fréchet-differentiable on $\mathcal{P}_2(\mathbb{R}^N)$, with first variation $\delta\mathcal{R}/\delta\rho$ well-defined; $\nabla_x(\delta\mathcal{R}/\delta\rho)$ is locally bounded.*
4. ***(Rigid $\mathcal{L}_{\mathrm{Burg}}$ consistency.)*** *The Burgers consistency penalty $\mathcal{L}_{\mathrm{Burg}}$ is sufficiently strong that, in the infinite-capacity noiseless limit, the learned score field satisfies the viscous Burgers structure*
$$\partial_\tau s_\tau + 2\, s_\tau \cdot \nabla\, s_\tau = \Delta s_\tau. \tag{156}$$
Moreover, $p_\tau \in \mathcal{P}_2(\mathbb{R}^N)$ is smooth in (x, τ) and strictly positive ($p_\tau > 0$).
5. ***(PDE guidance as a gradient drift.)*** *In the continuous-time limit, the combined PDE constraint losses $\mathcal{L}_{\mathrm{ent}}$ (Kruzhkov entropy penalty) and $\mathcal{L}_{\mathrm{Burg}}$ (Burgers consistency) induce an additional guidance velocity $g_\tau(x)$ in the reverse ODE that converges to the spatial gradient of a functional. Specifically, the total effective velocity field of the EntroDiff reverse sampler is*
$$v_\tau(x) = -a(\tau)\Big[\nabla_x \log p_\tau(x) + \nabla_x V(x) + \lambda\, \nabla_x \frac{\delta\mathcal{R}}{\delta\rho}(p_\tau)(x)\Big], \tag{157}$$
where V is the potential of the target distribution ($\rho^\star \propto e^{-V}$). The guidance contributions ∇V and $\lambda\nabla(\delta\mathcal{R}/\delta\rho)$ arise from the gradient of the PDE losses with respect to the sample trajectory, which by design converge to these functional-gradient limits when the losses are driven to zero in training.

Then the density evolution of the reverse sampler satisfies

$$\partial_\tau p_\tau = \nabla \cdot \Big(a(\tau)\, p_\tau\, \nabla \frac{\delta\mathcal{F}}{\delta\rho}(p_\tau)\Big). \tag{158}$$

Equivalently, under the time reparameterization $\theta(\tau) = \int_0^\tau a(r)\, dr$, we have

$$\partial_\theta p_\theta = \nabla \cdot \Big(p_\theta\, \nabla \frac{\delta\mathcal{F}}{\delta p}(p_\theta)\Big), \tag{159}$$

which is precisely the W_2-gradient flow of $\mathcal{F}$. Consequently, by the JKO theory, a single time-discrete step takes the variational form

$$p^{n+1} = \arg\min_{p \in \mathcal{P}_2(\mathbb{R}^N)} \Big\{\mathcal{F}(p) + \frac{1}{2\Delta\theta}\, W_2^2(p, p^n)\Big\}. \tag{160}$$

Proof. **Step 1: Reverse sampler as a continuity equation.**

Let X_τ denote the particle trajectory of the reverse sampler. If its continuous-time limit satisfies the ODE

$$\frac{dX_\tau}{d\tau} = v_\tau(X_\tau), \tag{161}$$

then by the standard pushforward formula, the particle distribution $p_\tau = \mathrm{Law}(X_\tau)$ satisfies the continuity equation

$$\partial_\tau p_\tau + \nabla \cdot (p_\tau\, v_\tau) = 0. \tag{162}$$

Thus, identifying the velocity field v_τ is the central task.

For any test function $\varphi \in C_c^\infty(\mathbb{R}^N)$, the identity follows from the chain rule:

$$\begin{aligned}\frac{d}{d\tau}\int_{\mathbb{R}^N} \varphi(x)\, p_\tau(x)\, dx &= \frac{d}{d\tau}\,\mathbb{E}[\varphi(X_\tau)] \\ &= \mathbb{E}[\nabla\varphi(X_\tau)\cdot v_\tau(X_\tau)] \\ &= \int \nabla\varphi(x)\cdot v_\tau(x)\, p_\tau(x)\, dx.\end{aligned} \tag{163}$$

Integration by parts of the right-hand side yields

$$\frac{d}{d\tau}\int \varphi\, p_\tau\, dx = -\int \varphi(x)\, \nabla\cdot(p_\tau\, v_\tau)(x)\, dx, \tag{164}$$

and since φ is arbitrary, the continuity equation (162) holds in the sense of distributions (and pointwise under the smoothness of p_τ and v_τ).

Step 2: Score drift and PDE guidance — the total effective velocity.

We identify v_τ as the sum of two contributions.

(a) Score drift (from denoising score matching). Under Hypothesis 1 (exact score) and Hypothesis 4 (rigid $\mathcal{L}_{\mathrm{Burg}}$), the network score converges to the true score:

$$s_\theta(x,\tau) = \nabla_x \log p_\tau(x). \tag{165}$$

The probability flow ODE induced by score matching alone has the velocity

$$v_\tau^{\mathrm{score}}(x) = -a(\tau)\, s_\theta(x,\tau) = -a(\tau)\, \nabla_x \log p_\tau(x). \tag{166}$$

(b) PDE guidance drift (from constraint losses). The EntroDiff loss family includes PDE-constraint terms beyond denoising score matching: $\mathcal{L}_{\mathrm{ent}}$ (Kruzhkov entropy penalty) and $\mathcal{L}_{\mathrm{Burg}}$ (Burgers consistency). These losses are functionals of the generated trajectory that penalize deviations from physical admissibility. In the continuous-time and infinite-data limit, their gradients with respect to the sample position induce an additional drift $g_\tau(x)$.

By Hypothesis 5, as the constraint losses are driven to zero, this guidance drift converges to the gradient of the potential that defines the target distribution plus the variational gradient of the PDE-constraint functional:

$$g_\tau(x) = -\Big(\nabla_x V(x) + \lambda\, \nabla_x \frac{\delta\mathcal{R}}{\delta\rho}(p_\tau)(x)\Big). \tag{167}$$

Here $\nabla V = -\nabla \log \rho^\star$ is the drift toward the target $\rho^\star \propto e^{-V}$, and $\lambda\nabla(\delta\mathcal{R}/\delta\rho)$ is the PDE-constraint drift (see Step 4 for the construction of $\mathcal{R}$).

(c) Total effective velocity. Summing (166) and (167), and absorbing the common factor $-a(\tau)$,

$$v_\tau(x) = -a(\tau)\Big[\nabla_x \log p_\tau(x) + \nabla_x V(x) + \lambda\, \nabla_x \frac{\delta\mathcal{R}}{\delta\rho}(p_\tau)(x)\Big]. \tag{168}$$

Substituting into the continuity equation gives

$$\partial_\tau p_\tau = \nabla\cdot\Big(a(\tau)\, p_\tau\, \big[\nabla \log p_\tau + \nabla V + \lambda\nabla\tfrac{\delta\mathcal{R}}{\delta\rho}\big]\Big). \tag{169}$$

We now show that the bracketed quantity in (168) is exactly $\nabla(\delta\mathcal{F}/\delta\rho)$.

Step 3: First variation of the entropy term.

Consider the relative entropy

$$\mathrm{Ent}(p \mid \rho^\star) = \int p \, \log \frac{p}{\rho^\star} \, dx. \tag{170}$$

Take a perturbation $p_\varepsilon = p + \varepsilon\eta$ with $\int \eta \, dx = 0$. A first-order expansion gives

$$\frac{d}{d\varepsilon}\Big|_{\varepsilon=0} \mathrm{Ent}(p_\varepsilon \mid \rho^\star) = \int \eta(x) \Big(\log \frac{p(x)}{\rho^\star(x)} + 1 \Big) \, dx. \tag{171}$$

Hence the L^2 first variation is

$$\frac{\delta}{\delta p} \mathrm{Ent}(p \mid \rho^\star) = \log \frac{p}{\rho^\star} + 1 \tag{172}$$

(the additive constant does not affect the gradient flow). Taking the spatial gradient,

$$\nabla \frac{\delta}{\delta p} \mathrm{Ent}(p \mid \rho^\star) = \nabla \log p - \nabla \log \rho^\star. \tag{173}$$

Since $\rho^\star \propto e^{-V}$, we have $-\nabla \log \rho^\star = \nabla V$, and therefore

$$\nabla \frac{\delta}{\delta p} \mathrm{Ent}(p \mid \rho^\star) = \nabla \log p + \nabla V. \tag{174}$$

Step 4: PDE constraint as a differentiable functional.

The PDE constraint (beyond $\mathcal{L}_{\mathrm{Burg}}$) must be expressible as a density functional $\mathcal{R}(\rho)$. Under Hypothesis 3, $\mathcal{R}$ is Fréchet-differentiable on $\mathcal{P}_2$, so its first variation $\frac{\delta\mathcal{R}}{\delta\rho}$ exists and satisfies

$$\frac{d}{d\varepsilon}\Big|_{\varepsilon=0} \mathcal{R}(\rho_\varepsilon) = \int \frac{\delta\mathcal{R}}{\delta\rho}(\rho) \, \eta \, dx \tag{175}$$

for any zero-mass perturbation η.

In the Wasserstein setting, perturbations are generated by a transport field ξ via the pushforward:

$$T_\varepsilon(x) = x + \varepsilon \, \xi(x), \qquad \rho_\varepsilon = (T_\varepsilon)_\# \, \rho. \tag{176}$$

The corresponding variational formula is

$$\frac{d}{d\varepsilon}\Big|_{\varepsilon=0} \mathcal{R}\big((\mathrm{Id} + \varepsilon\xi)_\# \rho\big) = \int \nabla \frac{\delta\mathcal{R}}{\delta\rho}(\rho) \cdot \xi \, \rho \, dx. \tag{177}$$

Thus, the $\mathcal{R}$ gradient flow term contributes a drift $-\nabla(\delta\mathcal{R}/\delta\rho)$ in the velocity field (see (167)).

Step 5: Identification with the free energy gradient.

Combining the entropy and PDE-constraint terms, the free energy is

$$\mathcal{F}(\rho) = \mathrm{Ent}(\rho \mid \rho^\star) + \lambda \, \mathcal{R}(\rho). \tag{178}$$

From (174) and Step 4, its spatial variational gradient is

$$\nabla \frac{\delta\mathcal{F}}{\delta\rho} = \nabla \frac{\delta}{\delta\rho} \mathrm{Ent}(\rho \mid \rho^\star) + \lambda \, \nabla \frac{\delta\mathcal{R}}{\delta\rho} = \nabla \log \rho + \nabla V + \lambda \, \nabla \frac{\delta\mathcal{R}}{\delta\rho}. \tag{179}$$

Comparing (168) with (179), the total effective velocity is exactly

$$v_\tau(x) = -a(\tau) \, \nabla_x \frac{\delta\mathcal{F}}{\delta\rho}(p_\tau)(x). \tag{180}$$

Substituting into the continuity equation (Step 1),

$$\partial_\tau p_\tau + \nabla \cdot \Big(p_\tau \Big[-a(\tau) \, \nabla \frac{\delta\mathcal{F}}{\delta\rho}(p_\tau) \Big] \Big) = 0, \tag{181}$$

i.e.,

$$\partial_\tau p_\tau = \nabla \cdot \Big(a(\tau)\, p_\tau\, \nabla \frac{\delta \mathcal{F}}{\delta \rho}(p_\tau) \Big). \tag{182}$$

To eliminate the time-weight $a(\tau)$, define the reparameterized time

$$\theta(\tau) = \int_0^\tau a(r)\, dr. \tag{183}$$

Since $a(\tau) > 0$, θ is strictly increasing and locally invertible. By the chain rule,

$$\partial_\tau p_\tau = \frac{d\theta}{d\tau}\, \partial_\theta p_\theta = a(\tau)\, \partial_\theta p_\theta. \tag{184}$$

Cancelling $a(\tau)$ yields the canonical Wasserstein-2 gradient flow form

$$\partial_\theta p_\theta = \nabla \cdot \Big(p_\theta\, \nabla \frac{\delta \mathcal{F}}{\delta p}(p_\theta) \Big). \tag{185}$$

This completes the bridge from the EntroDiff reverse sampler to an exact Wasserstein gradient flow. □

Step 6: JKO discretization from the Wasserstein gradient flow.

We now show that the implicit Euler discretization of the above gradient flow is equivalent to the JKO variational scheme.

6.1 Implicit Euler discretization. Consider a uniform time grid with step $\Delta\theta$ (in reparameterized time). Given $p^n \approx p_{\theta_n}$, the implicit Euler discretization of the gradient flow is

$$\frac{p^{n+1} - p^n}{\Delta\theta} = \nabla \cdot \Big(p^{n+1}\, \nabla \frac{\delta \mathcal{F}}{\delta \rho}(p^{n+1}) \Big). \tag{186}$$

6.2 Proximal step analogy. In Euclidean space, the implicit Euler step for $\dot{x} = -\nabla F(x)$ is equivalent to the proximal minimization

$$x^{n+1} = \arg\min_x \Big\{ F(x) + \frac{1}{2\Delta t}\, \|x - x^n\|^2 \Big\}. \tag{187}$$

The Wasserstein analogue replaces the Euclidean distance with W_2 and the Euclidean gradient with the continuity equation structure. The claim is that

$$p^{n+1} = \arg\min_{p \in \mathcal{P}_2(\mathbb{R}^N)} \Big\{ \mathcal{F}(p) + \frac{1}{2\Delta\theta}\, W_2^2(p, p^n) \Big\} \tag{188}$$

is the JKO step corresponding to the implicit Euler discretization above.

6.3 Variational derivation. Define the augmented functional

$$\mathcal{J}(p) = \mathcal{F}(p) + \frac{1}{2\Delta\theta}\, W_2^2(p, p^n), \tag{189}$$

and let p^{n+1} be a minimizer. In Wasserstein space, admissible variations are generated by transport maps rather than additive perturbations. Define the perturbation

$$T_\varepsilon(x) = x + \varepsilon\, \xi(x), \qquad p_\varepsilon = (T_\varepsilon)_\#\, p^{n+1}, \tag{190}$$

where ξ is an arbitrary smooth, compactly supported vector field. Optimality requires

$$\frac{d}{d\varepsilon}\Big|_{\varepsilon=0} \mathcal{J}(p_\varepsilon) = 0. \tag{191}$$

Variation of $\mathcal{F}$. By the Wasserstein variational formula,

$$\frac{d}{d\varepsilon}\Big|_{\varepsilon=0} \mathcal{F}(p_\varepsilon) = \int \nabla \frac{\delta \mathcal{F}}{\delta \rho}(p^{n+1}) \cdot \xi\, p^{n+1}\, dx. \tag{192}$$

Variation of the W_2^2 term. Let π be the optimal transport plan from p^{n+1} to p^n, with Kantorovich potential ϕ. A classical result of optimal transport theory gives

$$\frac{\delta}{\delta\rho}\Big(\frac{1}{2}\,W_2^2(\rho, p^n)\Big) = \phi, \tag{193}$$

and therefore

$$\frac{d}{d\varepsilon}\Big|_{\varepsilon=0}\frac{1}{2}\,W_2^2(p_\varepsilon, p^n) = \int \nabla\phi\cdot\xi\; p^{n+1}\,dx. \tag{194}$$

First-order condition. Combining both terms,

$$\int\Big[\nabla\frac{\delta\mathcal{F}}{\delta\rho}(p^{n+1}) + \frac{1}{\Delta\theta}\,\nabla\phi\Big]\cdot\xi\; p^{n+1}\,dx = 0. \tag{195}$$

Since ξ is arbitrary, the integrand must vanish p^{n+1}-almost everywhere, yielding

$$\nabla\phi = -\Delta\theta\,\nabla\frac{\delta\mathcal{F}}{\delta\rho}(p^{n+1}). \tag{196}$$

Connection to the optimal transport map. The Brenier theorem gives the optimal transport map T from p^{n+1} to p^n as $T(x) = x - \nabla\phi(x)$. Substituting the expression for $\nabla\phi$,

$$T(x) = x + \Delta\theta\,\nabla\frac{\delta\mathcal{F}}{\delta\rho}(p^{n+1}). \tag{197}$$

Deriving the discrete continuity equation. By the pushforward relation $p^n = T_\# p^{n+1}$, we expand for small $\Delta\theta$ (using the Jacobian expansion of the pushforward):

$$p^n(x) = p^{n+1}(x) - \Delta\theta\,\nabla\cdot\Big(p^{n+1}\,\nabla\frac{\delta\mathcal{F}}{\delta\rho}(p^{n+1})\Big) + o(\Delta\theta). \tag{198}$$

Rearranging,

$$\frac{p^{n+1}-p^n}{\Delta\theta} = \nabla\cdot\Big(p^{n+1}\,\nabla\frac{\delta\mathcal{F}}{\delta\rho}(p^{n+1})\Big) + o(1). \tag{199}$$

Taking $\Delta\theta\to 0$, we recover exactly the implicit Euler discretization (186) of the Wasserstein gradient flow.

6.4 Conclusion of the JKO derivation. Thus, a minimizer p^{n+1} of the JKO functional satisfies the implicit Euler discretization of the Wasserstein gradient flow. The reverse sampler, whose density evolution was established in Steps 1–5 as the gradient flow of $\mathcal{F}$, therefore admits the JKO variational characterization in the discrete-time limit $\Delta\tau\to 0$. This completes the proof of Theorem 5. □

Compact Fokker–Planck expansion.

For completeness, expanding the gradient flow structurally: using $\nabla\log p_\theta = \nabla p_\theta/p_\theta$ and the explicit form of the entropy variation from Step 3, the gradient flow PDE becomes

$$\partial_\theta p_\theta = \Delta p_\theta + \nabla\cdot(p_\theta\nabla V) + \lambda\,\nabla\cdot\Big(p_\theta\,\nabla\frac{\delta\mathcal{R}}{\delta\rho}\Big), \tag{200}$$

exhibiting the standard decomposition into a diffusion term (Δp_θ), an external potential drift ($\nabla\cdot(p_\theta\nabla V)$), and a PDE-constraint transport term.

Remark (Relation to the Obukhov–Fokker–Planck correspondence).

The identification (180) can be understood through the lens of entropy-dissipation balances. The pure reverse probability-flow ODE ($v = -a\nabla\log p$) is the Wasserstein gradient flow of the negative entropy $H(p) = \int p\log p$. Adding the PDE guidance terms $\nabla V + \lambda\nabla(\delta\mathcal{R}/\delta\rho)$ shifts the stationary point from the standard Gaussian to the target $\rho^\star$ while incorporating the physical constraint $\mathcal{R}$. The resulting free energy $\mathcal{F} = \mathrm{Ent}(\cdot\mid\rho^\star) + \lambda\mathcal{R}$ is the natural Lyapunov functional for the guided dynamics, and the JKO scheme is its canonical time discretization [8].

B Additional Experiments

B.1 E4 · 2D Shallow-water (rebuttal)

TODO: If time permits before rebuttal: hydraulic jump benchmark.

B.2 E5 · 1D Vlasov–Poisson (rebuttal)

TODO: Cold-beam instability with weak-solution filamentation. Compare against Deep Kinetic JKO [16].

B.3 Extended ablations

TODO: Ablate noise schedule, parameterization, loss-family components, and λ hyperparameters individually.

C Implementation Details

C.1 Network architecture

All experiments use a 1D U-Net backbone with four down/up-sampling stages and GroupNorm (8 groups). The baseline and StandardScore variants use base dimension $\mathrm{dim} = 64$ ($\approx 0.4\,\mathrm{M}$ parameters); the BV-aware variant (Eq. 4) uses $\mathrm{dim} = 128$ ($\approx 7.7\,\mathrm{M}$ parameters). The architecture follows the EDM preconditioning (Karras et al., 2022): $D_\theta(x, \sigma) = c_{\mathrm{skip}}(\sigma)\, x + c_{\mathrm{out}}(\sigma)\, F_\theta(c_{\mathrm{in}}(\sigma)\, x,\ c_{\mathrm{noise}}(\sigma))$, with $\sigma_{\mathrm{data}} = 0.5$. The BV-aware parameterization decomposes the score into three sub-networks: $\phi_\theta^{\mathrm{sm}}$ (the U-Net described above), $\phi_\theta^{\mathrm{sh}}$ (two Conv1d layers, 32 channels, kernel size 3, SiLU activation), and κ_θ (two Conv1d layers, 16 channels, kernel size 1, SiLU, with a Softplus gate enforcing $\kappa_\theta > 0$). The Tweedie inversion $D_x = x + \sigma^2 s_\theta$ is applied at the output to maintain compatibility with the EDM loss and sampler.

C.2 Training

All models are trained for the stated number of epochs (200 for BV-aware, 50 for StandardScore and Baseline) using the Adam optimiser with learning rate 2×10^{-4}, batch size 64, no weight decay, and a fixed random seed (PyTorch default; $\mathrm{seed} = 0$ for data generation). The viscosity-matched schedule sets $\nu_{\mathrm{phys}} = 1.0$, $T_d = 1.0$, giving $\sigma(\tau) = \sqrt{2\tau}$ with $\sigma \in [0, \sqrt{2}]$. The EDM baseline uses the standard log-normal schedule $\ln \sigma \sim \mathcal{N}(-1.2,\ 1.44)$. Loss weights are $\lambda_{\mathrm{DSM}} = 1.0$, $\lambda_{\mathrm{BV}} = 0.1$ for all EntroDiff configurations; the time-consistency term (Section 5.3) uses $\lambda_{\mathrm{time}} = 1.0$. Training was performed on a single NVIDIA GeForce GPU with 24 GB memory (RTX 3090/4090 class); one full run of the BV-aware configuration (200 epochs) completes in approximately 35 minutes.

C.3 Sampling

The Heun second-order integrator (Algorithm 1) uses $N_\tau = 50$ steps as default, with ablations at $N_\tau \in \{10, 25, 50, 100\}$. The Godunov-form PDE guidance coefficient ζ_{PDE} is set to 0 for unconditional generation; an ablation on $\zeta_{\mathrm{PDE}} \in \{0, 0.1, 0.5, 1.0\}$ (Figure A3 in Appendix B) confirms that the guidance term is numerically stable across all tested values but, for unconditional sampling, does not improve W_1 beyond the architectural prior, since the $\mathcal{L}_{\mathrm{PDE}}^{\mathrm{Godunov}}$ minimiser at unconstrained initial conditions is the trivial solution $u \equiv 0$. The time-step of the Godunov residual matches the data generation: $\Delta t = 5 \times 10^{-3}$, $\Delta x = 2\pi/128$. The guidance gradient is clipped to unit norm per spatial channel to prevent overflow from the large Godunov residuals of random initial noise.

C.4 Compute and reproducibility

All experiments were conducted on a Linux workstation with three NVIDIA GPUs (24 GB each), 125 GB RAM, and an NVMe SSD. Data generation (5,000 trajectories of 1D Burgers at $N_x = 128$) completes in under two minutes. The full experimental pipeline, including data generation, training,

and evaluation scripts, is available in the anonymised supplementary material under the MIT license. Key hyperparameters are summarised in Table 6.

Table 6: Hyperparameter summary.

Parameter	Value
Optimiser	Adam
Learning rate	2×10^{-4}
Batch size	64
ν_{phys} (schedule)	1.0
T_d	1.0
λ_{BV}	0.1
Heun steps N_τ	50
Data samples	5,000
Spatial grid N_x	128
Temporal steps N_t	100
Physical time T	0.5

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Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
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